$(c-1)$-BASIS FOR KBSM OF $\mathbf{A}^2 \times S^1$

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Abstract. In this paper we find a family of new basis for the Kauffman Bracket Skein Module (KBSM) of a product of annulus $\mathbf{A}^2$ and S^1, i.e., $\mathbf{A}^2 \times S^1$. This new family is a modification of the bases for $\mathbf{A}^2 \times S^1$ found previously by Dabkowski and Wu in [1]. The bases presented here can further be used to construct new families of bases for the fibered torus $V(\beta, 3)$. Such bases are needed while computing the KBSM of small Seifert fibered 3-manifolds with at least one fiber of multiplicity three. Thus, this work is a necessary step toward computing the KBSM of such manifolds. Furthermore, we present a Python-implemented algorithm for the change of basis, which facilitates computations in the KBSM for $\mathbf{A}^2 \times S^1$.

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1. Introduction

Skein modules for 3-manifolds were introduced by J. H. Przytycki in 1987 (see [6]). These algebraic objects are invariants of 3-manifolds that also allow us to study links in them. The simplest skein module is based on the Kauffman Bracket skein relation [4] for framed links in S^3, and; therefore, we call it the Kauffman Bracket Skein Module (KBSM). In the early papers, J. H. Przytycki [7] found KBSM for several families of 3-manifolds. The main techniques used to find them are combinatorial, i.e., based on diagrams of links on a surface. Using such techniques, KBSM for the family of 3 manifolds that are products of an oriented surface F and interval I, i.e., $F \times I$, were first determined. As shown by J. H. Przytycki, KBSM of such manifolds are free R-modules. In particular, for 3-manifold $M = T^2 \times I$, KBSM is a free R-module with a basis of simple closed framed curves and their parallel copies on T^2 including the empty link. In 2008, Dabkowski and Mroczkowski computed the KBSM of $F \times S^1$, where F is a compact planar surface F with 3 boundary components in [5] using a different basis for KBSM of $\mathbf{A}^2 \times S^1$. The main idea in [5] is to represent links via their arrow diagrams on $\mathbf{A}^2$ rather than classical diagrams on T^2. Furthermore, Dabkowski and Wu in [1] found a different basis for KBSM of $\mathbf{A}^2 \times S^1$. This family of bases was used to find an infinite family of bases for the KBSM of a $(\beta, 2)$-fibered torus in [2]. This, in turn, allowed them to compute KBSM of small Seifert fibered 3-manifolds with singular fibers of multiplicity two in [1]. To find KBSM for families of small

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Seifert fibered 3-manifolds with a singular fiber of multiplicity three, we need to find a basis for $(\beta, 3)$-fibered torus $V(\beta, 3)$. In this paper, we modify the basis

$$\Sigma_c = \{\lambda^n, x_c\lambda^n(x_c)^k, x_{c+1}\lambda^n(x_c)^k \mid \ n, k \geq 0\}$$

for the KBSM of $\mathbf{A}^2 \times S^1$ found in [1] to construct another infinite family of bases

$$\Sigma_{c,-} = \{\lambda^n, x_c\lambda^n(x_c)^k, x_{c-1}\lambda^n(x_c)^k \mid \ n, k \geq 0\},$$

that could next be used to determine a basis for $V(\beta, 3)$. This shows, in particular, that KBSM of $\mathbf{A}^2 \times S^1$ is the free R-module $R\Sigma_{c,-}$. Our notation with a negative sign reflects that we use $c-1$ instead of $c+1$. Modifying Σ_c into $\Sigma_{c,-}$ appears as a small change, there is no obvious change of basis, and it is not clear that $\Sigma_{c,-}$ is a basis for KBSM of $\mathbf{A}^2 \times S^1$. Therefore, the algorithm reducing any arrow diagram D in $\mathbf{A}^2$ to an R-linear combination of elements of Σ_c needs to be modified so that the new algorithm results in R-linear combination of $\Sigma_{c,-}$. Furthermore, we need to show that when we use arrow diagram D' obtained from D by a finite sequence of $\Omega_1 - \Omega_5$ moves then the resulting R-linear combinations are the same. The new family of bases for the KBSM of $\mathbf{A}^2 \times S^1$ found in this paper might be regarded as a first step toward finding bases for the KBSM of $(\beta, 3)$-fibered torus, where $\gcd(\beta, 3) = 1$ as well as a part of the larger goal of finding KBSM for small Seifert fibered 3-manifolds.

To make this paper largely self-contained, we recall the preliminary setup and terminology that will be used. As the major ideas for all the proofs are similar to those in [1], constant reference will be made to the mirrored proofs. The structure of the paper will begin with preliminary material (e.g., the definition of KBSM, description of links and their isotopies via arrow diagrams, etc.), and then a proof that $\Sigma_{c,-}$ is a basis for the KBSM of $\mathbf{A}^2 \times S^1$ will be given.

2. Preliminary

2.1. **Kauffman Bracket Skein Module.**

The definition of the Kauffman bracket skein module (KBSM) for an oriented 3-manifold M is motivated by relations used to define a regular isotopy invariant of links in S^3 called Kauffman Bracket polynomial $\langle \ \ \rangle \in \mathbb{Z}[A^{\pm 1}]$. Recall, in [4], L. H. Kauffman defined the bracket polynomial (now the Kauffman bracket) as follows:

Definition 2.1. [4] Let D be a link diagram. Let $\langle D \rangle$ be the element of the ring $\mathbb{Z}[A^{\pm 1}]$ defined using the following rules:

(1) $\langle \bigcirc \rangle = 1$
(2) $\langle \bigcirc \sqcup D \rangle = (-A^2 - A^{-2})\langle D \rangle$
(3) $\langle \times \rangle = A\langle \asymp \rangle + A^{-1}\langle \| \rangle$

The choice of coefficients in the above definition is not arbitrary. Indeed, Kauffman's choice of coefficients makes $\langle \ \ \rangle$ invariant for the second and third Reidemeister moves [4]. We also note that it is currently common to also consider the empty link $\emptyset$ and define $\langle \emptyset \rangle = 1$. Therefore, using (2), we see that

$$\langle \bigcirc \rangle = \langle \bigcirc \sqcup \emptyset \rangle = (-A^{-2} - A^2)\langle \emptyset \rangle = -A^{-2} - A^2.$$

and use this instead of (1). We can compute Kauffman bracket polynomial with modified versions of (1)-(3); however, for theoretical purposes, it is easier to use its definition as a state sum.

A *Kauffman state* is an assignment of a positive or negative marker to every crossing of a diagram D. Hence, for a diagram with n crossings, there are 2^n different Kauffman states s. Let $\mathcal{K}(D)$ be the set of all Kauffman states s of diagram D. Smoothing of a crossing changes the diagram only in a local disk around the crossing and depends on the marker as shown in Figure 2.1.

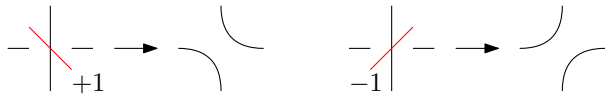


Figure 2.1. Smoothing crossing according to positive and negative markers

For a Kauffman state $s \in \mathcal{K}(D)$, let D_s be the diagram that results from smoothing all crossings according to the Kauffman state s. An example is shown in Figure 2.2.

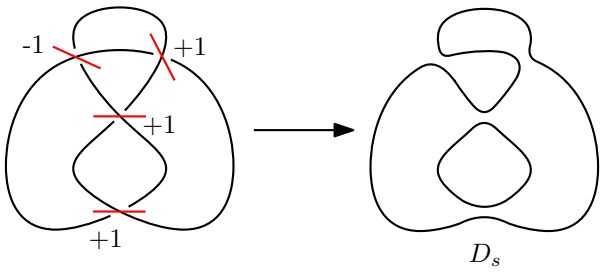


FIGURE 2.2. D_s obtained from D by smoothing its crossings according to s

Let $|D_s|$ be the number of trivial circles in D_s, and $p(s)$ and $n(s)$ be the number of positive and negative markers of s. For diagram D and state s in Figure 2.2, $|D_s| = 2$, $p(s) = 3$ and $n(s) = 1$.

The Kauffman bracket polynomial found with modified versions of (1)-(3) is equivalent to the polynomial found with the following state-sum formula.

Definition 2.2. Let D be a diagram in $\mathbb{R}^2$,

$$\langle D \rangle = \sum_{s \in \mathcal{K}(D)} A^{p(s)-n(s)}(-A^{-2} - A^2)^{|D_s|} \tag{1}$$

To return to the original state-sum formula in Kauffman's paper with condition (1) unmodified, subtract one from $|D_s|$ [4]. This formula is easier to work with than the recursive rule definition for theoretical purposes.

Using this modification of conditions (1)–(3) in the definition of the Kauffman bracket polynomial, we are ready to define the main object that we study in this paper, i.e., KBSM of an oriented 3-manifold M. We define *Kauffman bracket skein module* using an equivalent version of the definition given originally by J. H. Przytycki in [7].

Definition 2.3. Let M be an oriented 3-mainfold. Let R be a commutative ring with identity and an invertible element $A \in R$. Let $R\mathcal{L}^{fr}$ be the free R module with basis $\mathcal{L}^{fr}$ consisting of ambient isotopy classes of framed links in M (including the empty framed link). Let T_1 be the trivial framed knot. Let $S_{2,\infty}$ be the submodule of $R\mathcal{L}^{fr}$ generated by

$$L_+ - AL_0 - A^{-1}L_\infty \quad \text{and} \quad L \sqcup T_1 + (A^{-2} + A^2)L, \tag{2}$$

where framed links L_+, L_0, L_∞ are identical outside of a 3-ball $B^3 \subseteq M$ and differ inside of B^3 as shown in Figure 2.3, and $L \sqcup T_1$ is the disjoint union of a framed link L and the trivial framed knot T_1[1].

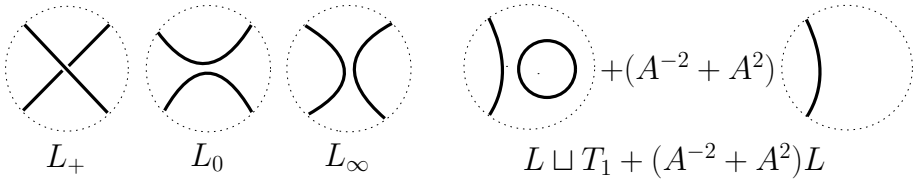


FIGURE 2.3. Skein triple L_+, L_0, L_∞ and relation $L \sqcup T_1 + (A^{-2} + A^2)L$

The *Kauffman bracket skein module* (KBSM), denoted as $\mathcal{S}_{2,\infty}(M; R, A)$, of M is defined as the quotient module of $R\mathcal{L}^{fr}$ by $S_{2,\infty}$, i.e.,

$$\mathcal{S}_{2,\infty}(M; R, A) = R\mathcal{L}^{fr}/S_{2,\infty}.$$

[1]There is a 3-ball B^3 in M such that $T_1 \subseteq B^3$ and $L \cap B^3 = \emptyset$

2.2. Ambient Isotopy of Framed Links in $\mathbf{A}^2 \times S^1$.

$\mathbf{A}^2 \times S^1$ is like the solid torus $\mathbf{D}^2 \times S^1$, but with an inner disk removed from the disk $\mathbf{D}^2$ to form the annulus $\mathbf{A}^2$ (see Figure 2.4).

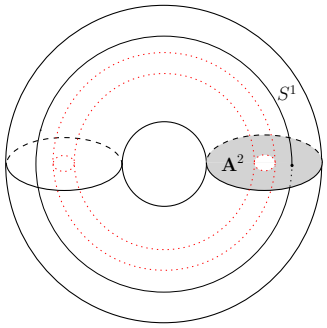


FIGURE 2.4. $\mathbf{A}^2 \times S^1$

A knot is a smooth embedding of the circle S^1 in $\mathbf{A}^2 \times S^1$. A link is a smooth embedding of a disjoint union of circles (i.e., disjoint union knots). A framed link is a smooth embedding of a disjoint family of annuli into $\mathbf{A}^2 \times S^1$. Let $p : \mathbf{A}^2 \times S^1 \to \mathbf{A}^2$ be a projection. A link L with a framing such that $p(L)$ is a curve on $\mathbf{A}^2$ will be called the standard framed link. As it can be argued, every framed link in $\mathbf{A}^2 \times S^1$ is ambient isotopic to a link with a standard framing. Therefore, we assume that all framed links have the standard framing. Any link in $\mathbf{A}^2 \times S^1$ with standard framing can be represented as an arrow diagram using the map in Figure 2.5 (see also [1]).

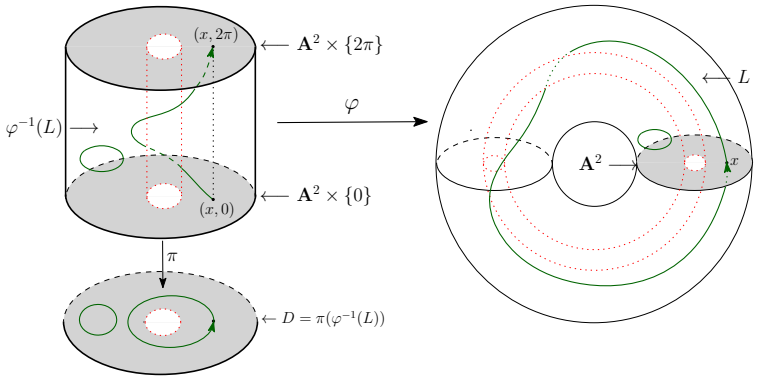


FIGURE 2.5. Arrow Diagram of a link in $\mathbf{A}^2 \times S^1$

It follows from [3] that two framed links in $\mathbf{A}^2 \times S^1$ are related by ambient isotopy if and only if their arrow diagrams are related by a composition of a finite number of five Reidemeister-type moves $\Omega_1 - \Omega_5$ (see Figure 2.6).

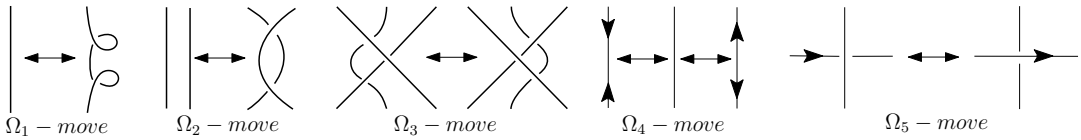


FIGURE 2.6. Arrow moves Ω_1, Ω_2, Ω_3, Ω_4, and Ω_5

When drawing arrow diagrams on $\mathbf{A}^2$, for convenience, we represent the inner circle of $\mathbf{A}^2$ as a point. This results in the representation of framed links in a 2-disk $D^2_\bullet$ with a marked point (see an arrow diagram in Figure 2.7).

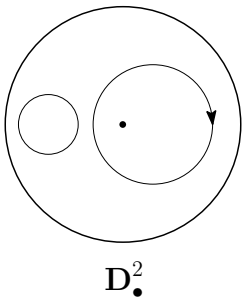


FIGURE 2.7. Arrow Diagram in Figure 2.5 on $D^2_\bullet$

In an arrow diagram, there may be many arrows along a curve between two consecutive crossings. To simplify such a diagram we may assign to each such curve a counterclockwise arrow with an integer. This integer is positive when the difference between the number of counterclockwise and clockwise arrows is positive, and negative otherwise. If the difference is zero, we do not place an arrow (see Figure 2.8).

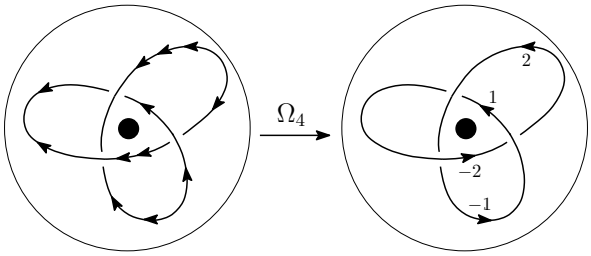


FIGURE 2.8. Example of reducing arrows in an arrow diagram to a tally

Now, we may study arrow diagrams on $\mathbf{A}^2$ up to Ω_1–Ω_5 moves, instead of framed links in $\mathbf{A}^2 \times S^1$ up to ambient isotopy.

3. Basis $\Sigma_{c,-}$ for KBSM of $\mathbf{A}^2 \times S^1$

3.1. Arrow Diagrams on 2-Disk $\mathbf{D}^2$.

Let $R = \mathbb{Z}[A^{\pm 1}]$. Let D be an arrow diagram on 2-disk, and we denote by $\mathcal{D}(\mathbf{D}^2)$ the set of all such diagrams. Let λ be a curve on 2-disk, and λ^n be n disjoint copies of λ (see Figure 3.1).

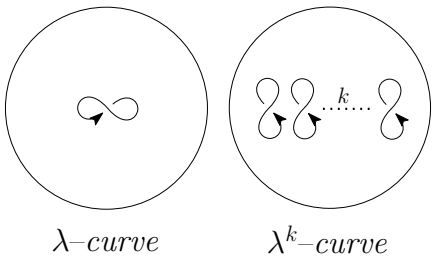


FIGURE 3.1. λ curve and λ^n curves

Let Λ be the set of all λ^n for every $n \geq 0$. In [5], the refined Kauffman bracket $\langle \cdot \rangle_r$ (see Definition 3.5) was defined. This $\langle \cdot \rangle_r$ gives an isomorphism of the free R-module of arrow diagrams on $\mathbf{D}^2$ up to $\Omega_1 - \Omega_5$-moves and the free R-module $R\Lambda$.

$$\langle \cdot \rangle_r : R\mathcal{D}(\mathbf{D}^2) \to R\Lambda$$

Proposition 3.7 from [5] shows that $\langle\cdot\rangle_r$ is invariant under $\Omega_2 - \Omega_5$-moves on the disk $\mathbf{D}^2$. We show that the refined bracket from [5] is also invariant under Ω_1-move to complete the argument[2].

Lemma 3.1. *For D and D' related by Ω_1 moves, $\langle D\rangle_r = \langle D'\rangle_r$.*

Proof. It suffices to show for a single Ω_1-move, and assume the diagram is identical outside of the disk where the Ω_1 move is performed as shown in Figure 3.2.

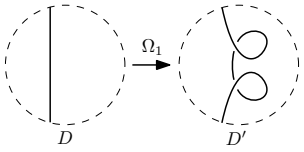


FIGURE 3.2. Diagrams D and D' related by an Ω_1 move.

From Figure 3.3 we can see that,

$$\langle D'\rangle_r = A^2(-A^{-2}-A^2)\langle D\rangle_r + \langle D\rangle_r + (-A^{-2}-A^2)^2\langle D\rangle_r + A^{-2}(-A^{-2}-A^2)\langle D\rangle_r.$$

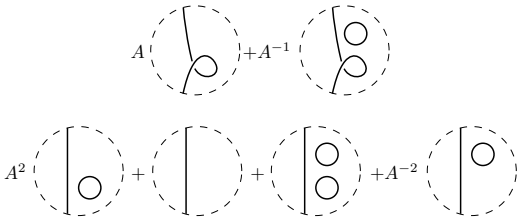


FIGURE 3.3. $\langle D'\rangle_r$ relationship to $\langle D\rangle_r$.

Hence,

$$\begin{aligned}\langle D'\rangle_r &= (A^2(-A^{-2}-A^2) + 1 + (-A^{-2}-A^2)^2 + A^{-2}(-A^{-2}-A^2))\langle D\rangle_r \\ &= (1 + (-A^{-2}-A^2)^2 - (-A^{-2}-A^2)(-A^{-2}-A^2))\langle D\rangle_r \\ &= \langle D\rangle_r,\end{aligned}$$

which completes our proof. □

The following are definitions and identities from [1] and [5] needed for the proofs in this paper.

Let t_n be a circle with n arrows, $n \in \mathbb{Z}$, and let $t_{n,k}$ be t_n with k disjoint λ curves inside as in Figure 3.4.

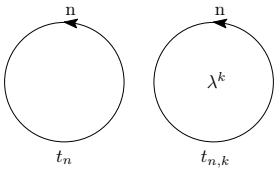


FIGURE 3.4. t_n and $t_{n,k}$.

In the KBSM of $\mathbf{D}^2 \times S^1$,

$$t_1 = -A^3\lambda, \quad t_{-1} = -A^{-3}\lambda,$$

and

$$t_n = A\lambda t_{n-1} - A^2 t_{n-2},$$

and

$$t_{n,k} = At_{n+1,k-1} + A^{-1}t_{n-1,k-1}.$$

[2]It is rather a common practice to omit Ω_1 - move since it is easy to show that Kauffman bracket is invariant under this move.

Definition 3.2 ([1])**.** A family of polynomials $\{P_n\}_{n\in\mathbb{Z}}$ in $R[\lambda]$ is determined by $P_0 = -A^2 - A^{-2}$, $P_1 = -A^3\lambda$, and the relation

$$P_n - A\lambda P_{n-1} + A^2 P_{n-2} = 0. \tag{3}$$

Definition 3.3 ([1])**.** A family of polynomials $\{P_{n,k}\}_{n\in\mathbb{Z},k\geq 0}$ in $R[\lambda]$ is determined by $P_{n,0} = P_n$, and for $k \geq 1$,

$$P_{n,k} = AP_{n+1,k-1} + A^{-1}P_{n-1,k-1}. \tag{4}$$

By the definition of $\langle\cdot\rangle_r$ in [5],

$$\langle t_n\rangle_r = P_n \quad \text{and} \quad \langle t_{n,k}\rangle_r = P_{n,k}.$$

It is shown in Figure 3.3 of [1] that the following identity holds

$$P_{n,k} = A\lambda P_{n-1,k} - A^2 P_{n-2,k}. \tag{5}$$

Definition 3.4 ([1])**.** A family of polynomials $\{Q_n\}_{n\in\mathbb{Z}}$ in $R[\lambda]$ is determined by recursion

$$Q_0 = 0, \quad Q_1 = 1, \quad \text{and} \quad Q_{n+2} = \lambda Q_{n+1} - Q_n$$

for $n \geq 0$, and $Q_n = -Q_{-n}$ for $n < 0$.

We will prove one of the identities from [1] that we need in this paper as an exercise.

Lemma 3.5. *For any $n \in \mathbb{Z}$,*

$$P_n = -A^{n+2}Q_{n+1} + A^{n-2}Q_{n-1}. \tag{6}$$

Proof. By Definition 3.2 and Definition 3.4,

$$P_0 = -A^2 - A^{-2} = -A^2Q_1 + A^{-2}Q_{-1} \quad \text{and} \quad P_1 = -A^3\lambda = -A^3Q_2 + A^{-1}Q_0.$$

If $n > 1$, assume P_n and P_{n+1} satisfy equation (6). Using equation (3) and Definition 3.4,

$$\begin{aligned} P_{n+2} &= A\lambda P_{n+1} - A^2 P_n \\ &= A\lambda(-A^{n+3}Q_{n+2} + A^{n-1}Q_n) - A^2(-A^{n+2}Q_{n+1} + A^{n-2}Q_{n-1}) \\ &= -A^{n+4}(\lambda Q_{n+2} - Q_{n+1}) + A^n(\lambda Q_n - Q_{n-1}) \\ &= -A^{n+4}Q_{n+3} + A^n Q_{n+1}. \end{aligned}$$

If $n < 0$, assume P_n and P_{n-1} satisfy equation (6). Using equation (3) and Definition 3.4,

$$\begin{aligned} P_{n-2} &= A^{-1}\lambda P_{n-1} - A^{-2}P_n \\ &= A^{-1}\lambda(-A^{n+1}Q_n + A^{n-3}Q_{n-2}) - A^{-2}(-A^{n+2}Q_{n+1} + A^{n-2}Q_{n-1}) \\ &= -A^n(\lambda Q_n - Q_{n+1}) + A^{n-4}(\lambda Q_{n-2} - Q_{n-1}) \\ &= -A^n Q_{n-1} + A^{n-4}Q_{n-3}. \end{aligned}$$

Thus, equation (6) follows by induction for all $n \in \mathbb{Z}$. This concludes the recounting of definitions and identities from [1]. □

For figures and more explanation, see [1] and [5]. These definitions and equations (3), (4), (5), and (6) will be used throughout the proofs of this paper. The annulus differs from the disk as curves can encircle the removed inner disk (marked point on $\mathbf{D}^2$). The idea used in [5] and reapplied in [1]– as well as here–is to use $\langle\cdot\rangle_r$ on the diagrams in between each of the curves that encircle the marked point.

3.2. KBSM of Arrow Diagrams on Annulus $\mathbf{A}^2$.

We will use the Kauffman state sum to express arrow diagrams as R-linear combination of arrow diagrams with no crossings. Let D be an arrow diagram with no crossings. The diagram can be written as a word in the following way described in [1]. Curves either encircle the center, which we will call x-curves, or they do not, in which case they will form a sub-diagram D_i on $\mathbf{D}^2$ (still with no crossings) in between x-curves. For diagram D_i, let i enumerate the sub-diagrams from in to out. For x-curve x_{m_i}, let i enumerate the curves from in to out, and m_i be the tally of arrows on the ith curve. For x-curves with the same tally of arrows and empty diagrams in between, we can write $(x_m)^k$ as k copies of the same curve (see Figure 3.5).

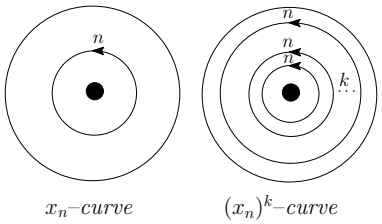


FIGURE 3.5. $x_n - curve$ and $(x_n)^k - curves$

For an arrow diagram D on $\mathbf{A}^2$ with no crossings shown in Figure 3.6,

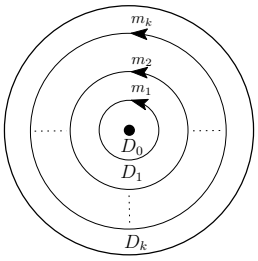


FIGURE 3.6. Arrow diagram D with no crossings written as a word

we write it as a word

$$D = D_0 x_{m_1} D_1 \dots D_{k-1} x_{m_k} D_k$$

Fix $c \in \mathbb{Z}$, let

$$\Sigma_{c,-} = \{\lambda^n, x_c \lambda^n (x_c)^k, x_{c-1} \lambda^n (x_c)^k \mid \ n, k \geq 0\}.$$

Let $\overline{\mathcal{D}}(\mathbf{A}^2)$ be the set of equivalence classes up to $\Omega_1 - \Omega_5$-moves of arrow diagrams on $\mathbf{A}^2$, and let $\overline{\Sigma}_{c,-}$ be the set of equivalence classes up to $\Omega_1 - \Omega_5$-moves of arrow diagrams in $\Sigma_{c,-}$. The relations (2) for arrow diagrams in $\mathbf{A}^2$ can be written as

$$[D_+] - A[D_0] - A^{-1}[D_\infty] \quad \text{and} \quad [D \sqcup T_1] + (A^{-2} + A^2)[D], \tag{7}$$

with ambient isotopy classes of framed links L, L_+, L_0, and L_∞ replaced by equivalence classes of corresponding arrow diagrams D, D_+, D_0, and D_∞ modulo $\Omega_1 - \Omega_5$-moves. Let $S_{2,\infty}(\mathbf{A}^2)$ be the submodule of $R\overline{\mathcal{D}}(\mathbf{A}^2)$ generated by relations (7). The KBSM of arrow diagrams on $\mathbf{A}^2 \times S^1$ is the quotient R-module

$$R\overline{\mathcal{D}}(\mathbf{A}^2)/S_{2,\infty}(\mathbf{A}^2).$$

In the next sections, we will show that $R\overline{\Sigma}_{c,-}$ is isomorphic to $R\overline{\mathcal{D}}(\mathbf{A}^2)/S_{2,\infty}(\mathbf{A}^2)$.

3.3. Surjective homomorphism h of free R-modules.

In [1], the surjective homomorphism h from $R\mathcal{D}(\mathbf{A}^2)$ to $R\Sigma_c$ is defined in three parts: h_1, h_2, and h_3 where

$$h : R\mathcal{D}(\mathbf{A}^2) \to R\Sigma_c \quad \text{and} \quad h = h_3 \circ h_2 \circ h_1.$$

In this paper, h_1 and h_2 are identical, and h_3 is modified. First, we recall the definitions of h_1 and h_2 from [1]. We denote by $\mathcal{D}(\mathbf{A}^2)$ the set of all arrow diagrams on $\mathbf{A}^2$, and by $\mathcal{D}'(\mathbf{A}^2)$ the set of all arrow diagrams in $\mathbf{A}^2$ with neither trivial circles nor crossings. For a given diagram $D \in \mathcal{D}(\mathbf{A}^2)$ and Kauffman states $s \in \mathcal{K}(D)$, we can define Kauffman state sum as given in Definition 2.2. Namely, we denote by D_s the diagram D smoothed according to state s; $D(s)$ – the diagram D after removing all trivial circles, $|D_s|$ – the number of trivial circles in D_s, and $p(s)$ and $n(s)$ – the number of positive and negative markers of s.

Definition 3.6. Let $R\mathcal{D}(\mathbf{A}^2)$ be the free R-module with basis of arrow diagrams on $\mathbf{A}^2$ and $R\mathcal{D}'(\mathbf{A}^2)$ be free R-module with basis of arrow diagrams with no trivial components and no crossings. We define homomorphism of free R-modules

$$h_1 : R\mathcal{D}(\mathbf{A}^2) \to R\mathcal{D}'(\mathbf{A}^2)$$

on generators by

$$h_1(D) = \sum_{s \in \mathcal{K}(D)} A^{p(s)-n(s)}(-A^{-2}-A^2)^{|D_s|}D(s). \tag{8}$$

The map h_1 is the Kauffman state-sum formula 1 for arrow diagrams on $\mathbf{A}^2$ instead of classical diagrams on $\mathbb{R}^2$. With classical diagrams on $\mathbb{R}^2$, $D(s)$ is the empty diagram because all components of D_s are trivial circles. This also holds true for non-arrow diagrams in the disk $\mathbf{D}^2$. This map turns an arrow diagram D into an R-linear combination of arrow diagrams with no crossings. As noted in [1], an arrow diagram $D \in \mathcal{D}'(\mathbf{A}^2)$ with no crossings and no trivial circle is still an arrow diagram $D \in \mathcal{D}(\mathbf{A}^2)$. Furthermore, for an arrow diagram with no crossings

$$h_1(D) = A^{0-0}(-A^{-2}-A^2)^0 D(s) = D(s) = D,$$

so h_1 is clearly surjective. Also,

$$h_1(D_+ - AD_0 - A^{-1}D_\infty) = h_1(D \sqcup T_1 + (A^{-2}+A^2)D) = 0,$$

and for D and D' related by $\Omega_1 - \Omega_4$ moves $h_1(D) = h_1(D')$, or $h_1(D - D') = 0$.

Let $R\Gamma$ be the free R-module with basis

$$\Gamma = \{\lambda^{n_0} x_{m_1} \lambda^{n_1} \cdots \lambda^{n_{k-1}} x_{m_k} \lambda^{n_k} \mid n_i \geq 0,\ m_i \in \mathbb{Z} \text{ and } k \geq 0\},$$

where $\lambda^{n_0} x_{m_1} \lambda^{n_1} \cdots \lambda^{n_{k-1}} x_{m_k} \lambda^{n_k}$ is shown in Figure 3.7.

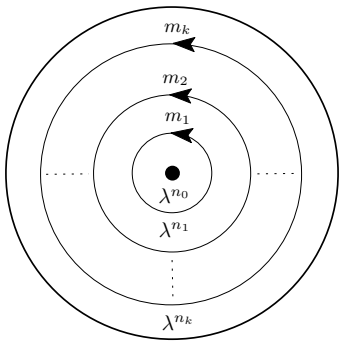


FIGURE 3.7. Generator $\lambda^{n_0} x_{m_1} \lambda^{n_1} \cdots \lambda^{n_{k-1}} x_{m_k} \lambda^{n_k}$ of KBSM of $\mathbf{A}^2 \times S^1$

This generating set Γ is similar to the generating set found in [5], except the λ curves are replaced with $-A^{-3}t_1$ curves and in [5] they are called x-type components. As in Figure 3.6, a diagram $D \in \mathcal{D}'(\mathbf{A}^2)$ with no crossings can be written as

$$D = D_0 x_{m_1} D_1 \ldots D_{k-1} x_{m_k} D_k.$$

For $D \in \mathcal{D}'(\mathbf{A}^2)$, let

$$\langle\!\langle D \rangle\!\rangle_\Gamma = \langle D_0 \rangle_r\, x_{m_1} \langle D_1 \rangle_r \ldots \langle D_{k-1} \rangle_r\, x_{m_k} \langle D_k \rangle_r.$$

Recall $\langle \cdot \rangle_r$ is the refined Kauffman bracket from [5], $\langle \cdot \rangle_r$ is invariant under $\Omega_1 - \Omega_5$ moves, and it returns elements of $R[\lambda]$.

Definition 3.7. Let $R\mathcal{D}'(\mathbf{A}^2)$ be a free R-module defined as above and $R\Gamma$ be free R-module with basis Γ. Define homomorphism of the modules

$$h_2 : R\mathcal{D}'(\mathbf{A}^2) \to R\Gamma$$

on generators by

$$h_2(\sum_{i=1}^{m} r_i w_i) = \sum_{i=1}^{m} r_i \langle\!\langle w_i \rangle\!\rangle_\Gamma \tag{9}$$

Since $\lambda = -A^3 t_1$, we may view each arrow diagram in Γ as a diagram with no crossings and trivial components thus, as noted in [1], for $D \in \Gamma$, $h_2(D) = D$, so h_2 is surjective. The construction of h_1 and h_2 is now complete. Now, proofs and results from [1] begin to diverge due to modification of h_3.

Lemma 3.8. *Let $c \in \mathbb{Z}$ and $w \in \Gamma$, then in KBSM of arrow diagrams the following relations hold*

(1) for $w = w'x_m\lambda^n(x_c)^k$ *with* w' *containing at least one* x *curve and* $n \geq 1$,

$$w = Aw'x_{m-1}\lambda^{n-1}(x_c)^k + A^{-1}w'x_{m+1}\lambda^{n-1}(x_c)^k;$$

(2) for $w = w'x_m(x_c)^k \in \Gamma$ *with* w' *containing at least one* x *curve and* $m > c$,

$$w = A^{-1}w'\lambda x_{m-1}(x_c)^k - A^{-2}w'x_{m-2}(x_c)^k;$$

(3) for $w = w'x_m(x_c)^k \in \Gamma$ *with* w' *containing at least one* x *curve and* $m < c-1$,

$$w = Aw'\lambda x_{m+1}(x_c)^k - A^2w'x_{m+2}(x_c)^k;$$

(4) for $w = w'x_m\lambda^n x_{c-1}(x_c)^k \in \Gamma$,

$$w = -Aw'\lambda t_{c-m,n}(x_c)^k + 2w't_{c-m+1,n}(x_c)^k + A^2w'x_{m-1}\lambda^n(x_c)^{k+1};$$

(5) for $w = \lambda^n x_m\lambda^{n'}(x_c)^k \in \Gamma$ *with* $n \geq 1$,

$$w = A\lambda^{n-1}x_{m+1}\lambda^{n'}(x_c)^k + A^{-1}\lambda^{n-1}x_{m-1}\lambda^{n'}(x_c)^k;$$

(6) for $w = x_m\lambda^n(x_c)^k \in \Gamma$ *with* $m > c$,

$$w = Ax_{m-1}\lambda^{n+1}(x_c)^k - A^2x_{m-2}\lambda^n(x_c)^k;$$

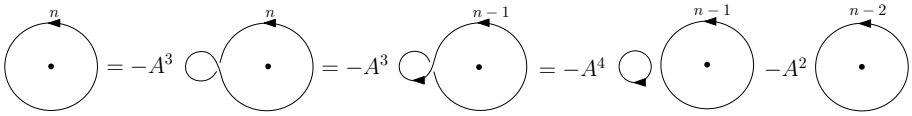
(7) for $w = x_m\lambda^n(x_c)^k \in \Gamma$ *with* $m < c-1$,

$$w = A^{-1}x_{m+1}\lambda^{n+1}(x_c)^k - A^{-2}x_{m+2}\lambda^n(x_c)^k.$$

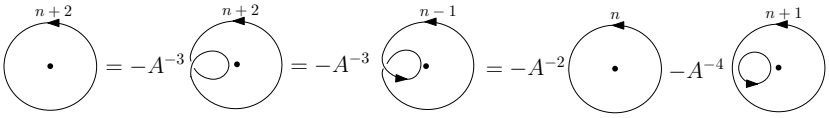
The proof of Lemma 3.8 mirrors Lemma 3.3 in [1] with the most notable difference in part (4).

Proof. As shown in Figure 3.8,

$$x_n = -A^4x_{n-1}t_{-1} - A^2x_{n-2} = -A^{-2}t_1x_{n+1} - A^2x_{n+2}.$$



$$x_n = -A^4x_{n-1}t_{-1} - A^2x_{n-2} = Ax_{n-1}\lambda - A^2x_{n-2}$$



$$x_{n+2} = -A^{-2}x_n - A^{-4}t_1x_{n+1} = -A^{-2}x_n + A^{-1}\lambda x_{n+1}$$

$$x_n = A\lambda x_{n+1} - A^2x_{n+2}$$

FIGURE 3.8. Relation for x_n-curve in $\mathbf{A}^2$

Substituting $t_{-1} = -A^{-3}\lambda$ and $t_1 = -A^3\lambda$, this becomes

$$x_n = A\lambda x_{n+1} - A^2x_{n+2} = Ax_{n-1}\lambda - A^2x_{n-2}. \tag{10}$$

Identity 1) begins with $w = w'x_m\lambda^n(x_c)^k$. By rearranging the right part of (10), it can become

$$x_{n-1}\lambda = Ax_{n-2} + A^{-1}x_n$$

Now this can be applied to $x_m\lambda$ in w.

$$w = w'x_m\lambda\lambda^{n-1}(x_c)^k = Aw'x_{m-1}\lambda^{n-1}(x_c)^k + A^{-1}w'x_{m+1}\lambda^{n-1}(x_c)^k$$

This is identity 1). Identity 2), 3), 5), 6), and 7) all follow similarly by applying (10). Identity 4) requires another relation.

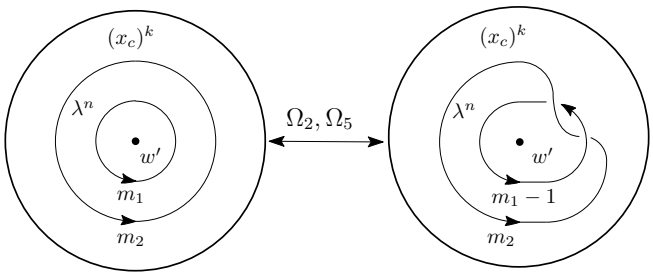


FIGURE 3.9. Ω_2 and Ω_5 moves between x_{m_1} and x_{m_2} curves on $\mathbf{A}^2$

The links in Figure 3.9 differ by Ω_2 and Ω_5 moves. Hence,

$$w'x_{m_1}\lambda^n x_{m_2}(x_c)^k = -Aw'\lambda t_{m_2-m_1+1,n}(x_n)^k + 2w't_{m_2-m_1+2,n}(x_n)^k + A^2w'x_{m_1-1}\lambda^n x_{m_2+1}(x_n)^k \quad (11)$$

If we take $m_2 = c-1$ in (11), it becomes identity 4).

$$w'x_m\lambda^n x_{c-1}(x_c)^k = -Aw'\lambda t_{c-m,n}(x_c)^k + 2w't_{c-m+1,n}(x_c)^k + A^2w'x_{m-1}\lambda^n(x_c)^{k+1}.$$

□

For a fixed $c \in \mathbb{Z}$, we consider a subset $\Sigma_{c,-}$ of Γ defined by

$$\Sigma_{c,-} = \{\lambda^n, x_c\lambda^n(x_c)^k, x_{c-1}\lambda^n(x_c)^k \mid \ n,k \geq 0\}.$$

Following [1], we define accordingly $\langle\!\langle u \rangle\!\rangle_{c,-} \in R\Sigma_{c,-}$ by

(a) if $u = \sum_{w\in S} r_w w$, S is a finite subset of Γ with at least two elements, let

$$\langle\!\langle u \rangle\!\rangle_{c,-} = \sum_{w\in S} r_w \langle\!\langle w \rangle\!\rangle_{c,-};$$

(b) for $w \in \Sigma_{c,-}$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = w;$$

(c) for $w = w'x_m\lambda^n(x_c)^k$, where w' contains at least one x curve and $n \geq 1$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A\langle\!\langle w'x_{m-1}\lambda^{n-1}(x_c)^k \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w'x_{m+1}\lambda^{n-1}(x_c)^k \rangle\!\rangle_{c,-};$$

(d) for $w = w'x_m(x_c)^k$, where w' contains at least one x curve and $m > c$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A^{-1}\langle\!\langle w'\lambda x_{m-1}(x_c)^k \rangle\!\rangle_{c,-} - A^{-2}\langle\!\langle w'x_{m-2}(x_c)^k \rangle\!\rangle_{c,-};$$

(e) for $w = w'x_m(x_c)^k$, where w' contains at least one x curve and $m < c-1$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A\langle\!\langle w'\lambda x_{m+1}(x_c)^k \rangle\!\rangle_{c,-} - A^2\langle\!\langle w'x_{m+2}(x_c)^k \rangle\!\rangle_{c,-};$$

(f) for $w = w'x_m\lambda^n x_{c-1}(x_c)^k$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = -A\langle\!\langle w'\lambda P_{c-m,n}(x_c)^k \rangle\!\rangle_{c,-} + 2\langle\!\langle w'P_{c-m+1,n}(x_c)^k \rangle\!\rangle_{c,-} + A^2\langle\!\langle w'x_{m-1}\lambda^n(x_c)^{k+1} \rangle\!\rangle_{c,-};$$

(g) for $w = \lambda^n x_m \lambda^{n'}(x_c)^k$ with $n \geq 1$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A\langle\!\langle \lambda^{n-1}x_{m+1}\lambda^{n'}(x_c)^k \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle \lambda^{n-1}x_{m-1}\lambda^{n'}(x_c)^k \rangle\!\rangle_{c,-};$$

(h) for $w = x_m\lambda^n(x_c)^k$ with $m > c$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A\langle\!\langle x_{m-1}\lambda^{n+1}(x_c)^k \rangle\!\rangle_{c,-} - A^2\langle\!\langle x_{m-2}\lambda^n(x_c)^k \rangle\!\rangle_{c,-};$$

(i) for $w = x_m\lambda^n(x_c)^k$ with $m < c-1$,

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A^{-1}\langle\!\langle x_{m+1}\lambda^{n+1}(x_c)^k \rangle\!\rangle_{c,-} - A^{-2}\langle\!\langle x_{m+2}\lambda^n(x_c)^k \rangle\!\rangle_{c,-}.$$

The difference between this and the definition of $\langle\!\langle u \rangle\!\rangle_{c,-} \in R\Sigma_{c,-}$ in [1] is part (f) and the bounds for m in parts (d), (e), (h), and (i). Therefore proofs involving (f) will be the most modified, and the rest are almost identical.

Lemma 3.9. *For any* $w = w_1 x_m w_2 \in \Gamma$

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A\langle\!\langle w_1 \lambda x_{m+1} w_2 \rangle\!\rangle_{c,-} - A^2 \langle\!\langle w_1 x_{m+2} w_2 \rangle\!\rangle_{c,-} \tag{12}$$

and

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-1} \lambda w_2 \rangle\!\rangle_{c,-} - A^2 \langle\!\langle w_1 x_{m-2} w_2 \rangle\!\rangle_{c,-} \tag{13}$$

This proof mirrors Lemma 3.4 from [1].

Proof. It suffices to show (12) and (13) hold for four sub-cases, i.e., w_1 can either be λ^{n_1} or $w_1' x_{m_1} \lambda^{n_1}$; w_2 can either be $\lambda^n (x_c)^k$ or $\lambda^n x_{c-1} (x_c)^k$. This is enough because the parts c), d), e), and f) in the definition of $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$ can be used to express the terms in equations (12) and (13) as linear combinations of the those same respective terms, but with w_1 and w_2 as mentioned. This creates four sub-cases for $w \in \Gamma$.

i) $w = \lambda^{n_1} x_m \lambda^n (x_c)^k$, where $w_1 = \lambda^{n_1}$ and $w_2 = \lambda^n (x_c)^k$.
ii) $w = \lambda^{n_1} x_m \lambda^n x_{c-1} (x_c)^k$, where $w_1 = \lambda^{n_1}$ and $w_2 = \lambda^n x_{c-1} (x_c)^k$.
iii) $w = w_1' x_{m_1} \lambda^{n_1} x_m \lambda^n (x_c)^k$, where $w_1 = w_1' x_{m_1} \lambda^{n_1}$ and $w_2 = \lambda^n (x_c)^k$.
iv) $w = w_1' x_{m_1} \lambda^{n_1} x_m \lambda^n x_{c-1} (x_c)^k$, where $w_1 = w_1' x_{m_1} \lambda^{n_1}$ and $w_2 = \lambda^n x_{c-1} (x_c)^k$.

Case i): $w = \lambda^{n_1} x_m \lambda^n (x_c)^k$. Part g) of the definition of $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$ states:

$$\langle\!\langle \lambda^n x_m \lambda^{n'} (x_c)^k \rangle\!\rangle_{c,-} = A\langle\!\langle \lambda^{n-1} x_{m+1} \lambda^{n'} (x_c)^k \rangle\!\rangle_{c,-} + A^{-1} \langle\!\langle \lambda^{n-1} x_{m-1} \lambda^{n'} (x_c)^k \rangle\!\rangle_{c,-}.$$

This implies

$$\langle\!\langle \lambda^{n-1} x_{m-1} \lambda^{n'} (x_c)^k \rangle\!\rangle_{c,-} = A\langle\!\langle \lambda^n x_m \lambda^{n'} (x_c)^k \rangle\!\rangle_{c,-} - A^2 \langle\!\langle \lambda^{n-1} x_{m+1} \lambda^{n'} (x_c)^k \rangle\!\rangle_{c,-}.$$

Set x_{m-1} as x_m and λ^{n-1} as λ^{n_1} and $\lambda^{n'}$ as λ^n to get equation (12).

For (13), using inductively g), one shows that

$$\begin{aligned}
\langle\!\langle \lambda^{n_1} x_m \lambda^n (x_c)^k \rangle\!\rangle_{c,-} &= \sum_{i=0}^{n_1} A^{n_1-2i} \binom{n_1}{i} \langle\!\langle x_{m+n_1-2i} \lambda^n (x_c)^k \rangle\!\rangle_{c,-}, \\
\langle\!\langle \lambda^{n_1} x_{m-1} \lambda^{n+1} (x_c)^k \rangle\!\rangle_{c,-} &= \sum_{i=0}^{n_1} A^{n_1-2i} \binom{n_1}{i} \langle\!\langle x_{m-1+n_1-2i} \lambda^{n+1} (x_c)^k \rangle\!\rangle_{c,-}, \\
\langle\!\langle \lambda^{n_1} x_{m-2} \lambda^n (x_c)^k \rangle\!\rangle_{c,-} &= \sum_{i=0}^{n_1} A^{n_1-2i} \binom{n_1}{i} \langle\!\langle x_{m-2+n_1-2i} \lambda^n (x_c)^k \rangle\!\rangle_{c,-}.
\end{aligned}$$

Use identity h) or i), depending on whether $m - 1 + n_1 - 2i$ is greater or less than c, in the definition of $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$. If $m - 1 + n_1 - 2i = c$ apply h) or i) to one of the other terms. This gives

$$\langle\!\langle x_{m-1+n_1-2i} \lambda^{n+1} (x_c)^k \rangle\!\rangle_{c,-} = A\langle\!\langle x_{m-2+n_1-2i} \lambda^n (x_c)^k \rangle\!\rangle_{c,-} + A^{-1} \langle\!\langle x_{m+n_1-2i} \lambda^n (x_c)^k \rangle\!\rangle_{c,-}$$

for all $i \in \mathbb{Z}$, in particular $i = 0, 1, ..., n_1$. This means we can show the following:

$$\begin{aligned}
\langle\!\langle \lambda^{n_1} x_{m-1} \lambda^{n+1} (x_c)^k \rangle\!\rangle_{c,-} &= \sum_{i=0}^{n_1} A^{n_1-2i} \binom{n_1}{i} \langle\!\langle x_{m-1+n_1-2i} \lambda^{n+1} (x_c)^k \rangle\!\rangle_{c,-} \\
&= \sum_{i=0}^{n_1} A^{n_1-2i} \binom{n_1}{i} \left(A\langle\!\langle x_{m-2+n_1-2i} \lambda^n (x_c)^k \rangle\!\rangle_{c,-} + A^{-1} \langle\!\langle x_{m+n_1-2i} \lambda^n (x_c)^k \rangle\!\rangle_{c,-} \right) \\
&= A\langle\!\langle \lambda^{n_1} x_{m-2} \lambda^n (x_c)^k \rangle\!\rangle_{c,-} + A^{-1} \langle\!\langle \lambda^{n_1} x_m \lambda^n (x_c)^k \rangle\!\rangle_{c,-}
\end{aligned}$$

Thus equation (13) follows from rearrangement.

Case ii): $w = \lambda^{n_1} x_m \lambda^n x_{c-1} (x_c)^k$. To show equation (12) use part f) in the definition of $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$.

$$\begin{aligned}
\langle\!\langle \lambda^{n_1} x_m \lambda^n x_{c-1}(x_c)^k \rangle\!\rangle_{c,-} &= -A\langle\!\langle \lambda^{n_1+1} P_{c-m,n}(x_c)^k \rangle\!\rangle_{c,-} + 2\langle\!\langle \lambda^{n_1} P_{c-m+1,n}(x_c)^k \rangle\!\rangle_{c,-} \\
&+ A^2\langle\!\langle \lambda^{n_1} x_{m-1}\lambda^n (x_c)^{k+1} \rangle\!\rangle_{c,-}, \\
\langle\!\langle \lambda^{n_1+1} x_{m+1} \lambda^n x_{c-1}(x_c)^k \rangle\!\rangle_{c,-} &= -A\langle\!\langle \lambda^{n_1+2} P_{c-m-1,n}(x_c)^k \rangle\!\rangle_{c,-} + 2\langle\!\langle \lambda^{n_1+1} P_{c-m,n}(x_c)^k \rangle\!\rangle_{c,-} \\
&+ A^2\langle\!\langle \lambda^{n_1+1} x_m\lambda^n (x_c)^{k+1} \rangle\!\rangle_{c,-}, \\
\langle\!\langle \lambda^{n_1} x_{m+2} \lambda^n x_{c-1}(x_c)^k \rangle\!\rangle_{c,-} &= -A\langle\!\langle \lambda^{n_1+1} P_{c-m-2,n}(x_c)^k \rangle\!\rangle_{c,-} + 2\langle\!\langle \lambda^{n_1} P_{c-m-1,n}(x_c)^k \rangle\!\rangle_{c,-} \\
&+ A^2\langle\!\langle \lambda^{n_1} x_{m+1}\lambda^n (x_c)^{k+1} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Using (4) to decompose the right hand side further and then Case i) it can be shown that

$$\langle\!\langle \lambda^{n_1} x_m \lambda^n x_{c-1}(x_c)^k \rangle\!\rangle_{c,-} = A\langle\!\langle \lambda^{n_1+1} x_{m+1}\lambda^n x_{c-1}(x_c)^k \rangle\!\rangle_{c,-} - A^2\langle\!\langle \lambda^{n_1} x_{m+2}\lambda^n x_{c-1}(x_c)^k \rangle\!\rangle_{c,-}$$

Case iii): $w = w_1' x_{m_1}\lambda^{n_1} x_m \lambda^n (x_c)^k$. Use part c) in the definition of $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$ on w we have

$$\langle\!\langle w \rangle\!\rangle_{c,-} = A\langle\!\langle w_1' x_{m_1}\lambda^{n_1} x_{m-1}\lambda^{n-1}(x_c)^k \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1' x_{m_1}\lambda^{n_1} x_{m+1}\lambda^{n-1}(x_c)^k \rangle\!\rangle_{c,-}.$$

Multiply by A and replace m with $m-1$ to get (13).
Following a similar argument as in Case i) for (12) we can use identity c) inductively.

$$\begin{aligned}
\langle\!\langle \lambda^{n_1} x_m \lambda^n (x_c)^k \rangle\!\rangle_{c,-} &= \sum_{i=0}^{n} A^{n-2i}\binom{n}{i}\langle\!\langle \lambda^{n_1} x_{m-n+2i}(x_c)^k \rangle\!\rangle_{c,-}, \\
\langle\!\langle \lambda^{n_1} \lambda x_{m+1} \lambda^n (x_c)^k \rangle\!\rangle_{c,-} &= \sum_{i=0}^{n} A^{n-2i}\binom{n}{i}\langle\!\langle \lambda^{n_1} \lambda x_{m+1-n+2i}(x_c)^k \rangle\!\rangle_{c,-}, \\
\langle\!\langle \lambda^{n_1} x_{m+2} \lambda^n (x_c)^k \rangle\!\rangle_{c,-} &= \sum_{i=0}^{n} A^{n-2i}\binom{n}{i}\langle\!\langle \lambda^{n_1} x_{m+2-n+2i}(x_c)^k \rangle\!\rangle_{c,-}.
\end{aligned}$$

For all $i = 0, 1, \ldots, n$, identity d) or e) gives,

$$\langle\!\langle \lambda^{n_1} \lambda x_{m+1-n+2i}(x_c)^k \rangle\!\rangle_{c,-} = A^{-1}\langle\!\langle \lambda^{n_1} x_{m-n+2i}(x_c)^k \rangle\!\rangle_{c,-} + A\langle\!\langle \lambda^{n_1} x_{m+2-n+2i}(x_c)^k \rangle\!\rangle_{c,-}.$$

Now, (12) follows from decomposing the summations and using the above equation to find equalities like in Case i).

Case iv): $w = w_1' x_{m_1}\lambda^{n_1} x_m \lambda^n x_{c-1}(x_c)^k$. Both equations (12) and (13) follow the argument in Case ii) exactly except $w_1 = w_1' x_{m_1}\lambda^{n_1}$ instead of $w_1 = \lambda^{n_1}$. The λ^{n_1} in the new w_1 will be changed in the same ways as in Case ii). □

Lemma 3.10. *For any $m, k \in \mathbb{Z}$ and $w_1 x_m w_2 \in \Gamma$*

$$\langle\!\langle w_1 x_m w_2 \rangle\!\rangle_{c,-} = -A^{m-k}\langle\!\langle w_1 x_k Q_{m-k-1} w_2 \rangle\!\rangle_{c,-} + A^{m-k-1}\langle\!\langle w_1 x_{k+1} Q_{m-k} w_2 \rangle\!\rangle_{c,-} \tag{14}$$

and

$$\langle\!\langle w_1 x_m w_2 \rangle\!\rangle_{c,-} = -A^{k-m}\langle\!\langle w_1 Q_{m-k-1} x_k w_2 \rangle\!\rangle_{c,-} + A^{k-m+1}\langle\!\langle w_1 Q_{m-k} x_{k+1} w_2 \rangle\!\rangle_{c,-}. \tag{15}$$

This proof mirrors Lemma 3.7 from [1].

Proof. For equation (14), if $m = k$,

$$\langle\!\langle w_1 x_k w_2 \rangle\!\rangle_{c,-} = -\langle\!\langle w_1 x_k Q_{-1} w_2 \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_{k+1} Q_0 w_2 \rangle\!\rangle_{c,-} = \langle\!\langle w_1 x_k w_2 \rangle\!\rangle_{c,-},$$

and if $m = k+1$,

$$\langle\!\langle w_1 x_k w_2 \rangle\!\rangle_{c,-} = -A\langle\!\langle w_1 x_k Q_0 w_2 \rangle\!\rangle_{c,-} + \langle\!\langle w_1 x_{k+1} Q_1 w_2 \rangle\!\rangle_{c,-} = \langle\!\langle w_1 x_k w_2 \rangle\!\rangle_{c,-}.$$

Let $m > k+1$ and assume equation (14) is true for $m-1$ and $m-2$. Using equation (13), and Definition 3.4, we see that

$$\begin{aligned}\langle\!\langle w_1 x_m w_2 \rangle\!\rangle_{c,-} &\overset{(13)}{=} A\langle\!\langle w_1 x_{m-1}\lambda w_2\rangle\!\rangle_{c,-} - A^2\langle\!\langle w_1 x_{m-2} w_2\rangle\!\rangle_{c,-}\\ &= -A^{m-k}\langle\!\langle w_1 x_k Q_{m-k-2}\lambda w_2\rangle\!\rangle_{c,-} + A^{m-k-1}\langle\!\langle w_1 x_{k+1} Q_{m-k-1}\lambda w_2\rangle\!\rangle_{c,-}\\ &+ A^{m-k}\langle\!\langle w_1 x_k Q_{m-k-3} w_2\rangle\!\rangle_{c,-} - A^{m-k-1}\langle\!\langle w_1 x_{k+1} Q_{m-k-2} w_2\rangle\!\rangle_{c,-}\\ &\overset{3.4}{=} -A^{m-k}\langle\!\langle w_1 x_{k+1} Q_{m-k-1} w_2\rangle\!\rangle_{c,-} + A^{m-k-1}\langle\!\langle w_1 x_{k+1} Q_{m-k} w_2\rangle\!\rangle_{c,-}.\end{aligned}$$

Let $m < k$ and assume equation (14) is true for $m+1$ and $m+2$. Using equation (13), and Definition 3.4, we see that

$$\begin{aligned}\langle\!\langle w_1 x_m w_2 \rangle\!\rangle_{c,-} &\overset{(13)}{=} A^{-1}\langle\!\langle w_1 x_{m+1}\lambda w_2\rangle\!\rangle_{c,-} - A^{-2}\langle\!\langle w_1 x_{m+2} w_2\rangle\!\rangle_{c,-}\\ &= -A^{m-k}\langle\!\langle w_1 x_k Q_{m-k}\lambda w_2\rangle\!\rangle_{c,-} + A^{m-k-1}\langle\!\langle w_1 x_{k+1} Q_{m-k+1}\lambda w_2\rangle\!\rangle_{c,-}\\ &+ A^{m-k}\langle\!\langle w_1 x_k Q_{m-k+1} w_2\rangle\!\rangle_{c,-} - A^{m-k-1}\langle\!\langle w_1 x_{k+1} Q_{m-k+2} w_2\rangle\!\rangle_{c,-}\\ &\overset{3.4}{=} -A^{m-k}\langle\!\langle w_1 x_{k+1} Q_{m-k-1} w_2\rangle\!\rangle_{c,-} + A^{m-k-1}\langle\!\langle w_1 x_{k+1} Q_{m-k} w_2\rangle\!\rangle_{c,-}.\end{aligned}$$

By induction, equation (14) is true for all $m \in \mathbb{Z}$. The induction for equation (15) follows similarly. □

Lemma 3.11. *Let $\Delta_t^+, \Delta_t^-, \Delta_x^+, \Delta_x^-$ be finite subsets of $\mathbb{Z}[A^{\pm}] \times \Gamma \times \Gamma \times \mathbb{Z}$, and define*

$$\Theta_t^+(k,n) = \sum_{(r,w_1,w_2,v)\in\Delta_t^+} r\langle\!\langle w_1 P_{n+v,k} w_2\rangle\!\rangle_{c,-}, \quad \Theta_t^-(k,n) = \sum_{(r,w_1,w_2,v)\in\Delta_t^-} r\langle\!\langle w_1 P_{-n+v}\lambda^k w_2\rangle\!\rangle_{c,-},$$

$$\Theta_x^+(k,n) = \sum_{(r,w_1,w_2,v)\in\Delta_x^+} r\langle\!\langle w_1 \lambda^k x_{n+v} w_2\rangle\!\rangle_{c,-}, \quad \Theta_x^-(k,n) = \sum_{(r,w_1,w_2,v)\in\Delta_x^-} r\langle\!\langle w_1 x_{-n+v}\lambda^k w_2\rangle\!\rangle_{c,-},$$

and

$$\Theta_{t,x}(k,n) = \Theta_t^+(k,n) + \Theta_t^-(k,n) + \Theta_x^+(k,n) + \Theta_x^-(k,n).$$

If either

1) $\Theta_{t,x}(0,n) = 0$ *for all* $n \in \mathbb{Z}$
   or
2) $\Theta_{t,x}(k,n_0) = \Theta_{t,x}(k,n_0+1) = 0$ *for all* $k \geq 0$ *and a fixed* $n_0 \in \mathbb{Z}$,
   then $\Theta_{t,x}(k,n) = 0$ *for any* $k \geq 0$ *and* $n \in \mathbb{Z}$.

This proof mirrors Lemma 3.8 from [1].

Proof. We prove 1) with induction on k. Assume $\Theta_{t,x}(0,n) = 0$ for all $n \in \mathbb{Z}$. This is the base case. Assume $\Theta_{t,x}(k-1,n) = 0$ for fixed $k \geq 1$. We need to show $\Theta_{t,x}(k,n) = 0$. Using (4), (5), (12), and (13) respectively,

$$\begin{aligned}\Theta_t^+(k,n) &= A\Theta_t^+(k-1,n+1) + A^{-1}\Theta_t^+(k-1,n-1),\\ \Theta_t^-(k,n) &= A\Theta_t^-(k-1,n+1) + A^{-1}\Theta_t^-(k-1,n-1),\\ \Theta_x^+(k,n) &= A\Theta_x^+(k-1,n+1) + A^{-1}\Theta_x^+(k-1,n-1),\\ \Theta_x^-(k,n) &= A\Theta_x^-(k-1,n+1) + A^{-1}\Theta_x^-(k-1,n-1).\end{aligned}$$

Thus we have

$$\Theta_{t,x}(k,n) = A\Theta_{t,x}(k-1,n+1) + A^{-1}\Theta_{t,x}(k-1,n-1).$$

By the inductive hypothesis $\Theta_{t,x}(k-1,n) = 0$ for all n, so $\Theta_{t,x}(k,n) = 0$ also.

We prove 2) with induction on n. Assume $\Theta_{t,x}(k,n_0) = \Theta_{t,x}(k,n_0+1) = 0$ for all $k \geq 0$ and a fixed $n_0 \in \mathbb{Z}$. This is the base case. Now we need to show

i) For $n > n_0 + 1$.
   If $\Theta_{t,x}(k,n-1) = \Theta_{t,x}(k,n-2) = 0$ for any k, then for $\Theta_{t,x}(k,n) = 0$ for any k.
ii) For $n < n_0$.
   If $\Theta_{t,x}(k,n+1) = \Theta_{t,x}(k,n+2) = 0$ for any k, then for $\Theta_{t,x}(k,n) = 0$ for any k.

For i), using (4), (5), (12), and (13) respectively,

$$\begin{aligned}
\Theta_t^+(k,n) &= A^{-1}\Theta_t^+(k+1,n-1) - A^{-2}\Theta_t^+(k,n-2),\\
\Theta_t^-(k,n) &= A^{-1}\Theta_t^-(k+1,n-1) - A^{-2}\Theta_t^-(k,n-2),\\
\Theta_x^+(k,n) &= A^{-1}\Theta_x^+(k+1,n-1) - A^{-2}\Theta_x^+(k,n-2),\\
\Theta_x^-(k,n) &= A^{-1}\Theta_x^-(k+1,n-1) - A^{-2}\Theta_x^-(k,n-2),
\end{aligned}$$

thus we have

$$\Theta_{t,x}(k,n) = A^{-1}\Theta_{t,x}(k+1,n-1) - A^{-2}\Theta_{t,x}(k,n-2).$$

By the induction hypothesis $\Theta_{t,x}(k,n-1) = \Theta_{t,x}(k,n-2) = 0$ for any k, so $\Theta_{t,x}(k,n) = 0$ also.

For ii), using (4), (5), (12), and (13) respectively,

$$\begin{aligned}
\Theta_t^+(k,n) &= A\Theta_t^+(k+1,n+1) - A^2\Theta_t^+(k,n+2),\\
\Theta_t^-(k,n) &= A\Theta_t^-(k+1,n+1) - A^2\Theta_t^-(k,n+2),\\
\Theta_x^+(k,n) &= A\Theta_x^+(k+1,n+1) - A^2\Theta_x^+(k,n+2),\\
\Theta_x^-(k,n) &= A\Theta_x^-(k+1,n+1) - A^2\Theta_x^-(k,n+2),
\end{aligned}$$

thus we have

$$\Theta_{t,x}(k,n) = A\Theta_{t,x}(k+1,n+1) - A^2\Theta_{t,x}(k,n+2).$$

By the induction hypothesis $\Theta_{t,x}(k,n+1) = \Theta_{t,x}(k,n+2) = 0$ for any k, so $\Theta_{t,x}(k,n) = 0$ also. □

Since $\Sigma_c \subset \Gamma$, in fact $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$ defines a surjective homomorphism of free R-modules

Definition 3.12. Let $R\Gamma$ be free R-module with basis Γ and $R\Sigma_{c,-}$ be the free R-module with basis $\Sigma_{c,-}$. Define homomorphism of the free R-modules

$$h_3 : R\Gamma \to R\Sigma_{c,-}$$

on generators as follows

$$h_3(u) = \langle\!\langle u \rangle\!\rangle_{c,-}.$$

Therefore, we can also define a surjective homomorphism of free R-modules

$$h : R\mathcal{D}(\mathbf{A}^2) \to R\Sigma_{c,-}$$

by $h(D) = (h_3 \circ h_2 \circ h_1)(D)$ for each generator $D \in \mathcal{D}(\mathbf{A}^2)$.

Example 3.13. We will perform a sample calculation on two arrow diagrams D and D' related by $\Omega_1 - \Omega_5$-moves in Figure 3.10 in the case when $c = 1$, so we take

$$\Sigma_{1,-} = \{\lambda^n, x_1\lambda^n(x_1)^k, x_0\lambda^n(x_1)^k \mid \ n,k \geq 0\}.$$

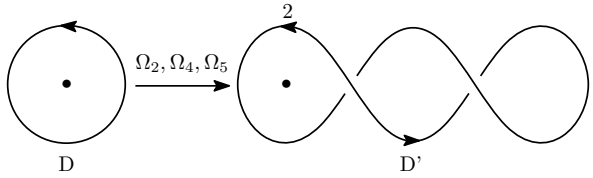


FIGURE 3.10. Arrow diagrams D and D' related by an Ω_2, Ω_4, and Ω_5 moves.

Calculating $h(D)$ is simple:

$$h(D) = h_3 \circ h_2 \circ h_1(D) = h_3 \circ h_2(x_1) = h_3(x_1) = x_1.$$

Diagram D was already in $\Sigma_{1,-}$, so it did not change. After performing Ω_2, Ω_4, and Ω_5 moves, diagram D becomes D', and the calculation becomes more complicated, but $h(D')$ should still be the same.

$$\begin{aligned}
h(D') &= h_3 \circ h_2 \circ h_1(D')\\
&= h_3 \circ h_2(A^2x_3 + x_2t_1 + (-A^2 - A^{-2})x_3 + A^{-2}(-A^2 - A^{-2})x_2t_1) = h_3 \circ h_2(-A^{-2}x_3 - A^{-4}x_2t_1)\\
&= h_3(-A^{-2}x_3 + A^{-1}x_2\lambda) = -A^{-2}(A^2x_1\lambda^2 - A^3x_0\lambda - A^2x_1) + A^{-1}(Ax_1\lambda - A^2x_0)\lambda\\
&= -x_1\lambda^2 + Ax_0\lambda + x_1 + x_1\lambda^2 - Ax_0\lambda = x_1.
\end{aligned}$$

Therefore, $h(D) = h(D')$. We note that this is not just a coincidence, since we will prove that, in fact, h is invariant under all compositions of finite number of $\Omega_1 - \Omega_5$-moves. Consequently, h is well-defined on equivalence classes of arrow diagrams on $\mathbf{A}^2$ modulo $\Omega_1 - \Omega_5$-moves.

3.4. **Invariance of h under $\Omega_1 - \Omega_5$-moves.**

We will show that for arrow diagrams D and D' related by an Ω_5-move, $h(D - D') = 0$. As the first step we consider special cases of arrow diagrams on $\mathbf{A}^2$ with one crossing which are related by Ω_5-move. Assume that $D = w_1 W w_2$, $D' = w_1 W' w_2$ with $w_1, w_2 \in \Gamma$ and W and W' related by an Ω_5-move. For such diagrams

$$h(D - D') = h(w_1(W - W')w_2) = h_3(w_1 h_2(A\langle W_+\rangle + A^{-1}\langle W_-\rangle - A\langle W'_+\rangle - A^{-1}\langle W'_-\rangle)w_2), \tag{16}$$

where W_+, W_-, W'_+, and W'_- are arrow diagrams obtained from W and W' by smoothing the crossing of each according to positive and negative markers. We must show that

$$h_3(w_1 h_2(A\langle W_+\rangle + A^{-1}\langle W_-\rangle - A\langle W'_+\rangle - A^{-1}\langle W'_-\rangle)w_2) = 0$$

for all special cases of arrow diagrams on $\mathbf{A}^2$ with one crossing which are related by Ω_5-move. Below are all the arrow diagrams with one crossing encircling the labeled point with λ^k only.

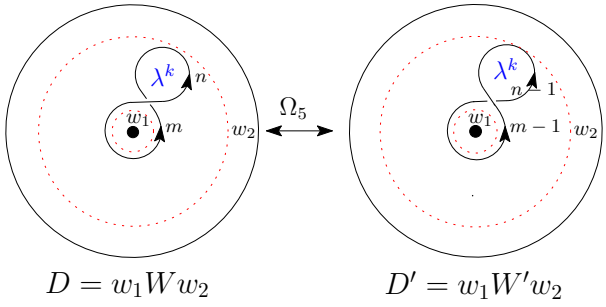


FIGURE 3.11. Lemma 3.15 Arrow diagrams D and D' related by an Ω_5-move on $\mathbf{A}^2$

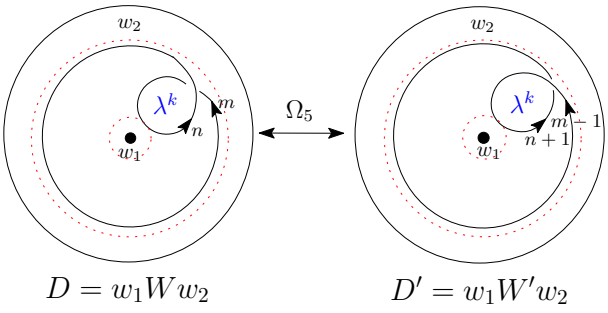


FIGURE 3.12. Lemma 3.16 Arrow diagrams D and D' related by an Ω_5-move on $\mathbf{A}^2$

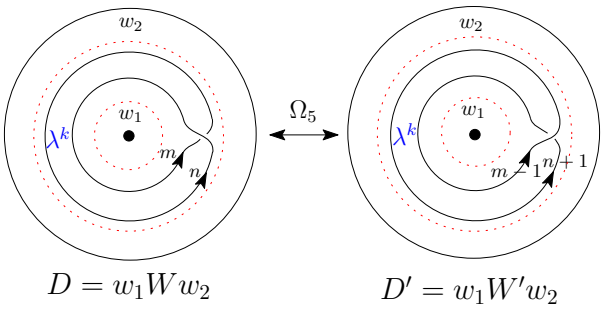


FIGURE 3.13. Lemma 3.17 Arrow diagrams D and D' related by an Ω_5-move on $\mathbf{A}^2$

For each of these diagrams D and D' related by an Ω_5-move we must show equation $h(D - D') = 0$.

For arrow diagrams D, D' in Figure 3.11, we see that $W_+ = \lambda^k x_{m+n}$, $W_- = x_m t_{n,k}$, $W'_+ = x_{m-1} t_{n-1,k}$, and $W'_- = \lambda^k x_{m+n-2}$ are obtained by smoothing the crossing of W and W' according to positive and negative markers. Hence, by (16), $h(D - D') = 0$ is equivalent to:

$$h_3(w_1(A\lambda^k x_{m+n} + A^{-1} x_m P_{n,k} - A x_{m-1} P_{n-1,k} - A^{-1}\lambda^k x_{m+n-2})w_2) = 0, \tag{17}$$

for all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$.

For arrow diagrams D, D' in Figure 3.12, we see that $W_+ = t_{n,k} x_m$, $W_- = x_{m-n}\lambda^k$, $W'_+ = x_{m-n-2}\lambda^k$, and $W'_- = t_{n+1,k} x_{m-1}$ are obtained by smoothing the crossing of W and W' according to positive and negative markers. Hence, by (16), $h(D - D') = 0$ is equivalent to:

$$h_3(w_1(A P_{n,k} x_m + A^{-1} x_{m-n}\lambda^k - A x_{m-n-2}\lambda^k - A^{-1} P_{n+1,k} x_{m-1})w_2) = 0, \tag{18}$$

for all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$.

For arrow diagrams D, D' in Figure 3.13, we see that $W_+ = t_{n-m,k}$, $W_- = x_m \lambda^k x_n$, $W'_+ = x_{m-1}\lambda^k x_{n+1}$, and $W'_- = t_{n-m+2,k}$ are obtained by smoothing the crossing of W and W' according to positive and negative markers. Hence, by (16), $h(D - D') = 0$ is equivalent to:

$$h_3(w_1(A P_{n-m,k} + A^{-1} x_m \lambda^k x_n - A x_{m-1}\lambda^k x_{n+1} - A^{-1} P_{n-m+2,k})w_2) = 0, \tag{19}$$

for all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$.

First, we will show the equality that will help simplify proof of Lemma 3.17.

Lemma 3.14. *For all $w_1 \in \Gamma$ and all $m, k \in \mathbb{Z}$ with $k \geq 0$,*

$$\langle\!\langle w_1 P_{c-m} x_{c-1} (x_c)^k \rangle\!\rangle_{c,-} - A^2 \langle\!\langle w_1 P_{c-m-1} (x_c)^{k+1} \rangle\!\rangle_{c,-} = -A^2 \langle\!\langle w_1 x_{m-1} (x_c)^k \rangle\!\rangle_{c,-} + \langle\!\langle w_1 x_{m+1} (x_c)^k \rangle\!\rangle_{c,-}. \tag{20}$$

This proof mirrors proof of Lemma 3.11 from [1].

Proof. Begin with the right hand side of equation (20). Use equation (6), the fact (*) $Q_n = -Q_{-n}$, and equation (15).

$$\begin{aligned}
&\langle\!\langle w_1 P_{c-m} x_{c-1} (x_c)^k \rangle\!\rangle_{c,-} - A^2 \langle\!\langle w_1 P_{c-m-1} (x_c)^{k+1} \rangle\!\rangle_{c,-} \\
&\overset{(6)}{=} -A^{c-m+2} \langle\!\langle w_1 Q_{c-m+1} x_{c-1} (x_c)^k \rangle\!\rangle_{c,-} + A^{c-m-2} \langle\!\langle w_1 Q_{c-m-1} x_{c-1} (x_c)^k \rangle\!\rangle_{c,-} \\
&+ A^{c-m+3} \langle\!\langle w_1 Q_{c-m} (x_c)^{k+1} \rangle\!\rangle_{c,-} - A^{c-m-1} \langle\!\langle w_1 Q_{c-m-2} (x_c)^{k+1} \rangle\!\rangle_{c,-} \\
&\overset{*}{=} -A^2(-A^{c-m} \langle\!\langle w_1 Q_{m-c-1} x_{c-1} (x_c)^k \rangle\!\rangle_{c,-} + A^{c-m+1} \langle\!\langle w_1 Q_{m-c} (x_c)^{k+1} \rangle\!\rangle_{c,-}) \\
&- A^{c-m-2} \langle\!\langle w_1 Q_{m-c+1} x_{c-1} (x_c)^k \rangle\!\rangle_{c,-} + A^{c-m-1} \langle\!\langle w_1 Q_{m-c+2} (x_c)^{k+1} \rangle\!\rangle_{c,-} \\
&\overset{(15)}{=} -A^2 \langle\!\langle w_1 x_{m-1} (x_c)^k \rangle\!\rangle_{c,-} + \langle\!\langle w_1 x_{m+1} (x_c)^k \rangle\!\rangle_{c,-}.
\end{aligned}$$

This gives the right hand side. □

Lemma 3.15. *For all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$,*

$$h_3(w_1(A\lambda^k x_{m+n} + A^{-1}x_m P_{n,k} - Ax_{m-1}P_{n-1,k} - A^{-1}\lambda^k x_{m+n-2})w_2) = 0. \tag{21}$$

which implies (17)*.*

This proof mirrors proof of Lemma 3.9 from [1].

Proof. If, for all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$,

$$A\langle\!\langle w_1\lambda^k x_{m+n}w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m P_{n,k} w_2\rangle\!\rangle_{c,-} - A\langle\!\langle w_1 x_{m-1}P_{n-1,k}w_2\rangle\!\rangle_{c,-} - A^{-1}\langle\!\langle w_1\lambda^k x_{m+n-2}w_2\rangle\!\rangle_{c,-} = 0. \tag{22}$$

is true, then (21) holds. Using part 1) of Lemma 3.11 we must show it is true for $k = 0$, and it will also hold for any $k \geq 0$. Setting $k = 0$ equation (22) is equivalent to

$$A\langle\!\langle w_1 x_{m+n}w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m P_n w_2\rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-1}P_{n-1}w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_{m+n-2}w_2\rangle\!\rangle_{c,-}. \tag{23}$$

Using equation (14) from Lemma 3.10, the first term becomes

$$A\langle\!\langle w_1 x_{m+n}w_2\rangle\!\rangle_{c,-} = A(-A^{n+1}\langle\!\langle w_1 x_{m-1}Q_n w_2\rangle\!\rangle_{c,-} + A^n\langle\!\langle w_1 x_m Q_{n+1}w_2\rangle\!\rangle_{c,-}),$$

and the fourth becomes

$$A^{-1}\langle\!\langle w_1 x_{m+n-2}w_2\rangle\!\rangle_{c,-} = A^{-1}(-A^{n-1}\langle\!\langle w_1 x_{m-1}Q_{n-2}w_2\rangle\!\rangle_{c,-} + A^{n-2}\langle\!\langle w_1 x_m Q_{n-1}w_2\rangle\!\rangle_{c,-}).$$

Using equation (6) from Lemma 3.5, the second term becomes

$$A^{-1}\langle\!\langle w_1 x_m P_n w_2\rangle\!\rangle_{c,-} = A^{-1}\langle\!\langle w_1 x_m(-A^{n+2}Q_{n+1} + A^{n-2}Q_{n-1})w_2\rangle\!\rangle_{c,-},$$

and the third becomes

$$A\langle\!\langle w_1 x_{m-1}P_{n-1}w_2\rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-1}(-A^{n+1}Q_n + A^{n-3}Q_{n-2})w_2\rangle\!\rangle_{c,-}.$$

Begin with the left hand side of (23).

$$\begin{aligned}
& A\langle\!\langle w_1 x_{m+n}w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m P_n w_2\rangle\!\rangle_{c,-} \\
= \;& A(-A^{n+1}\langle\!\langle w_1 x_{m-1}Q_n w_2\rangle\!\rangle_{c,-} + A^n\langle\!\langle w_1 x_m Q_{n+1}w_2\rangle\!\rangle_{c,-}) \\
+ \;& A^{-1}\langle\!\langle w_1 x_m(-A^{n+2}Q_{n+1} + A^{n-2}Q_{n-1})w_2\rangle\!\rangle_{c,-} = -A^{n+2}\langle\!\langle w_1 x_{m-1}Q_n w_2\rangle\!\rangle_{c,-} + A^{n+1}\langle\!\langle w_1 x_m Q_{n+1}w_2\rangle\!\rangle_{c,-} \\
- \;& A^{n+1}\langle\!\langle w_1 x_m Q_{n+1}w_2\rangle\!\rangle_{c,-} + A^{n-3}\langle\!\langle w_1 Q_{n-1}w_2\rangle\!\rangle_{c,-} = -A^{n+2}\langle\!\langle w_1 x_{m-1}Q_n w_2\rangle\!\rangle_{c,-} + A^{n-3}\langle\!\langle w_1 Q_{n-1}w_2\rangle\!\rangle_{c,-} \\
= \;& -A^{n+2}\langle\!\langle w_1 x_{m-1}Q_n w_2\rangle\!\rangle_{c,-} + A^{n-2}\langle\!\langle w_1 x_m Q_{n-2}w_2\rangle\!\rangle_{c,-} \\
- \;& A^{n-2}\langle\!\langle w_1 x_m Q_{n-2}w_2\rangle\!\rangle_{c,-} + A^{n-3}\langle\!\langle w_1 Q_{n-1}w_2\rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-1}(-A^{n+1}Q_n + A^{n-3}Q_{n-2})w_2\rangle\!\rangle_{c,-} \\
+ \;& A^{-1}(-A^{n-1}\langle\!\langle w_1 x_{m-1}Q_{n-2}w_2\rangle\!\rangle_{c,-} + A^{n-2}\langle\!\langle w_1 x_m Q_{n-1}w_2\rangle\!\rangle_{c,-}) \\
= \;& A\langle\!\langle w_1 x_{m-1}P_{n-1}w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_{m+n-2}w_2\rangle\!\rangle_{c,-}.
\end{aligned}$$

This is the right hand side of (23). □

Lemma 3.16. *For all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$,*

$$h_3(w_1(AP_{n,k}x_m + A^{-1}x_{m-n}\lambda^k - Ax_{m-n-2}\lambda^k - A^{-1}P_{n+1,k}x_{m-1})w_2) = 0. \tag{24}$$

which implies (18)*.*

This proof mirrors proof of Lemma 3.10 from [1].

Proof. If, for all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$,

$$A\langle\!\langle w_1 P_{n,k}x_m w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_{m-n}\lambda^k w_2\rangle\!\rangle_{c,-} - A\langle\!\langle w_1 x_{m-n-2}\lambda^k w_2\rangle\!\rangle_{c,-} - A^{-1}\langle\!\langle w_1 P_{n+1,k}x_{m-1}w_2\rangle\!\rangle_{c,-} = 0. \tag{25}$$

is true, then (24) holds. Using part 1) of Lemma 3.11 we must show it is true for $k = 0$, and it will also hold for any $k \geq 0$. Setting $k = 0$ equation (25) is equivalent to

$$A\langle\!\langle w_1 P_n x_m w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_{m-n}w_2\rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-n-2}w_2\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{n+1}x_{m-1}w_2\rangle\!\rangle_{c,-}. \tag{26}$$

Using equation (6) from Lemma 3.5, the first term becomes

$$A\langle\!\langle w_1 P_n x_m w_2\rangle\!\rangle_{c,-} = A\langle\!\langle w_1(-A^{n+2}Q_{n+1} + A^{n-2}Q_{n-1})x_m w_2\rangle\!\rangle_{c,-},$$

and the fourth becomes

$$A^{-1}\langle\!\langle w_1 P_{n+1} x_{m-1} w_2 \rangle\!\rangle_{c,-} = A^{-1}\langle\!\langle w_1(-A^{n+3}Q_{n+2} + A^{n-1}Q_n)x_{m-1}w_2\rangle\!\rangle_{c,-}.$$

Using equation (15) from Lemma 3.10, the second term becomes

$$A^{-1}\langle\!\langle w_1 x_{m-n} w_2 \rangle\!\rangle_{c,-} = A^{-1}(-A^{n-1}\langle\!\langle w_1 Q_{-n} x_{m-1} w_2 \rangle\!\rangle_{c,-} + A^n \langle\!\langle w_1 Q_{1-n} x_m w_2 \rangle\!\rangle_{c,-}),$$

and the third becomes

$$A\langle\!\langle w_1 x_{m-n-2} w_2 \rangle\!\rangle_{c,-} = A(-A^{n+1}\langle\!\langle w_1 Q_{-n-2} x_{m-1} w_2 \rangle\!\rangle_{c,-} + A^{n+2}\langle\!\langle w_1 Q_{-n-1} x_m w_2 \rangle\!\rangle_{c,-}).$$

Begin with the left hand side of (26).

$$\begin{aligned}
& A\langle\!\langle w_1 P_n x_m w_2 \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_{m-n} w_2 \rangle\!\rangle_{c,-} = A\langle\!\langle w_1(-A^{n+2}Q_{n+1} + A^{n-2}Q_{n-1})x_m w_2\rangle\!\rangle_{c,-} \\
+ \;& A^{-1}(-A^{n-1}\langle\!\langle w_1 Q_{-n} x_{m-1} w_2 \rangle\!\rangle_{c,-} + A^n \langle\!\langle w_1 Q_{1-n} x_m w_2 \rangle\!\rangle_{c,-}) \\
= \;& -A^{n+3}\langle\!\langle w_1 Q_{n+1} x_m w_2 \rangle\!\rangle_{c,-} + A^{n-1}\langle\!\langle w_1 Q_{n-1} x_m w_2 \rangle\!\rangle_{c,-} - A^{n-2}\langle\!\langle w_1 Q_{-n} x_{m-1} w_2 \rangle\!\rangle_{c,-} \\
+ \;& A^{n-1}\langle\!\langle w_1 Q_{1-n} x_m w_2 \rangle\!\rangle_{c,-} \overset{*}{=} -A^{n+3}\langle\!\langle w_1 Q_{n+1} x_m w_2 \rangle\!\rangle_{c,-} + A^{n-1}\langle\!\langle w_1 Q_{n-1} x_m w_2 \rangle\!\rangle_{c,-} \\
+ \;& A^{n-2}\langle\!\langle w_1 Q_n x_{m-1} w_2 \rangle\!\rangle_{c,-} - A^{n-1}\langle\!\langle w_1 Q_{n-1} x_m w_2 \rangle\!\rangle_{c,-} = -A^{n+3}\langle\!\langle w_1 Q_{n+1} x_m w_2 \rangle\!\rangle_{c,-} \\
+ \;& A^{n-2}\langle\!\langle w_1 Q_n x_{m-1} w_2 \rangle\!\rangle_{c,-} = -A^{n+3}\langle\!\langle w_1 Q_{n+1} x_m w_2 \rangle\!\rangle_{c,-} + A^{n+2}\langle\!\langle w_1 Q_{n+2} x_{m-1} w_2 \rangle\!\rangle_{c,-} \\
+ \;& A^{n-2}\langle\!\langle w_1 Q_n x_{m-1} w_2 \rangle\!\rangle_{c,-} - A^{n+2}\langle\!\langle w_1 Q_{n+2} x_{m-1} w_2 \rangle\!\rangle_{c,-} \overset{*}{=} A^{n+3}\langle\!\langle w_1 Q_{-n-1} x_m w_2 \rangle\!\rangle_{c,-} \\
- \;& A^{n+2}\langle\!\langle w_1 Q_{-n-2} x_{m-1} w_2 \rangle\!\rangle_{c,-} + A^{n-2}\langle\!\langle w_1 Q_n x_{m-1} w_2 \rangle\!\rangle_{c,-} - A^{n+2}\langle\!\langle w_1 Q_{n+2} x_{m-1} w_2 \rangle\!\rangle_{c,-} \\
= \;& A(-A^{n+1}\langle\!\langle w_1 Q_{-n-2} x_{m-1} w_2 \rangle\!\rangle_{c,-} + A^{n+2}\langle\!\langle w_1 Q_{-n-1} x_m w_2 \rangle\!\rangle_{c,-}) + A^{-1}\langle\!\langle w_1(-A^{n+3}Q_{n+2} \\
+ \;& A^{n-1}Q_n)x_{m-1}w_2\rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-n-2} w_2 \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{n+1} x_{m-1} w_2 \rangle\!\rangle_{c,-}.
\end{aligned}$$

Equality with "*" above use property $Q_n = -Q_{-n}$ which follows from definition 3.4. This is the right hand side of (26). □

Lemma 3.17. *For all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$,*

$$h_3(w_1(AP_{n-m,k} + A^{-1}x_m\lambda^k x_n - Ax_{m-1}\lambda^k x_{n+1} - A^{-1}P_{n-m+2,k})w_2) = 0. \tag{27}$$

which implies (19).

This proof mirrors proof of Lemma 3.12 from [1].

Proof. If, for all $w_1, w_2 \in \Gamma$, $m, n \in \mathbb{Z}$, and $k \geq 0$,

$$A\langle\!\langle w_1 P_{n-m,k} w_2 \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m \lambda^k x_n w_2 \rangle\!\rangle_{c,-} - A\langle\!\langle w_1 x_{m-1}\lambda^k x_{n+1} w_2 \rangle\!\rangle_{c,-} - A^{-1}\langle\!\langle w_1 P_{n-m+2,k} w_2 \rangle\!\rangle_{c,-} = 0. \tag{28}$$

is true, then (27) holds. Using part 1) of Lemma 3.11 we must show it is true for $k = 0$, and it will also hold for any $k \geq 0$. Setting $k = 0$ equation (28) is equivalent to

$$A\langle\!\langle w_1 P_{n-m} w_2 \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m x_n w_2 \rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-1} x_{n+1} w_2 \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{n-m+2} w_2 \rangle\!\rangle_{c,-}. \tag{29}$$

It suffices to show this is true for w_2 as either $\lambda^{n'}(x_c)^{k'}$ or $\lambda^{n'}x_{c-1}(x_c)^{k'}$. Using part 1) and 2) of Lemma 3.11 we must show equation (29) is true for $n' = 0$ and $n = c, c-1$. Then equation (29) will be true for all $n' \geq 0$ and $n \in \mathbb{Z}$. This creates four cases.

i) $n = c$ and $w_2 = (x_c)^{k'}$.

$$A\langle\!\langle w_1 P_{c-m}(x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m (x_c)^{k'+1} \rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-1} x_{c+1}(x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+2}(x_c)^{k'} \rangle\!\rangle_{c,-}.$$

ii) $n = c-1$ and $w_2 = (x_c)^{k'}$.

$$A\langle\!\langle w_1 P_{c-1-m}(x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m x_{c-1}(x_c)^{k'} \rangle\!\rangle_{c,-} = A\langle\!\langle w_1 x_{m-1}(x_c)^{k'+1} \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'} \rangle\!\rangle_{c,-}.$$

iii) $n = c$ and $w_2 = x_{c-1}(x_c)^{k'}$.

$$\begin{aligned}
& A\langle\!\langle w_1 P_{c-m} x_{c-1}(x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m x_c x_{c-1}(x_c)^{k'} \rangle\!\rangle_{c,-} = \\
& A\langle\!\langle w_1 x_{m-1} x_{c+1} x_{c-1}(x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+2} x_{c-1}(x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

iv) $n = c-1$ and $w_2 = x_{c-1}(x_c)^{k'}$

$$A\langle\!\langle w_1 P_{c-1-m} x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m x_{c-1} x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-} =$$
$$A\langle\!\langle w_1 x_{m-1} x_c x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+1} x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-}.$$

For Case i): Start with the right hand side. Use equation (12), part f) in Definition of $\langle\!\langle\cdot\rangle\!\rangle_{c,-}$, (3), and (13).

$$\begin{aligned}
&A\langle\!\langle w_1 x_{m-1} x_{c+1}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+2}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&\overset{(12)}{=} \langle\!\langle w_1 x_{m-1}\lambda(x_c)^{k'+1}\rangle\!\rangle_{c,-} - A^{-1}\langle\!\langle w_1 x_{m-1} x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+2}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&\overset{f}{=} \langle\!\langle w_1 x_{m-1}\lambda(x_c)^{k'+1}\rangle\!\rangle_{c,-} - A^{-1}\langle\!\langle w_1 P_{c-m+2}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&+ \langle\!\langle w_1\lambda P_{c-m+1}(x_c)^{k'}\rangle\!\rangle_{c,-} - A\langle\!\langle w_1 x_{m-2}(x_c)^{k'+1}\rangle\!\rangle_{c,-}\\
&\overset{(3)}{=} A\langle\!\langle w_1 P_{c-m}(x_c)^{k'}\rangle\!\rangle_{c,-} + \langle\!\langle w_1 x_{m-1}\lambda(x_c)^{k'+1}\rangle\!\rangle_{c,-} - A\langle\!\langle w_1 x_{m-2}(x_c)^{k'+1}\rangle\!\rangle_{c,-}\\
&\overset{(13)}{=} A\langle\!\langle w_1 P_{c-m}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m(x_c)^{k'+1}\rangle\!\rangle_{c,-}.
\end{aligned}$$

This is the left hand side.

For Case ii): Start with the left hand side. Use part f) in the definition of $\langle\!\langle\cdot\rangle\!\rangle_{c,-}$ and then equation (3).

$$\begin{aligned}
&A\langle\!\langle w_1 P_{c-1-m}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_m x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
\overset{f}{=}\ & A\langle\!\langle w_1 P_{c-1-m}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}(-A\langle\!\langle w_1\lambda P_{c-m}(x_c)^{k'}\rangle\!\rangle_{c,-} + 2\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
+\ & A^2\langle\!\langle w_1 x_{m-1}(x_c)^{k'+1}\rangle\!\rangle_{c,-}) = A\langle\!\langle w_1 x_{m-1}(x_c)^{k'+1}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
+\ & A^{-1}\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'}\rangle\!\rangle_{c,-} - \langle\!\langle w_1\lambda P_{c-m}(x_c)^{k'}\rangle\!\rangle_{c,-} + A\langle\!\langle w_1 P_{c-m-1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
\overset{(3)}{=}\ & A\langle\!\langle w_1 x_{m-1}(x_c)^{k'+1}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'}\rangle\!\rangle_{c,-}
\end{aligned}$$

This gives us the right hand side.

For Case iii) To prove, break down each term one at a time, and then combine.

Term 1: Use equation (20).

$$\begin{aligned}
&A\langle\!\langle w_1 P_{c-m} x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&\overset{(20)}{=} A^3\langle\!\langle w_1 P_{c-m-1}(x_c)^{k'+1}\rangle\!\rangle_{c,-} - A^3\langle\!\langle w_1 x_{m-1}(x_c)^{k'}\rangle\!\rangle_{c,-} + A\langle\!\langle w_1 x_{m+1}(x_c)^{k'}\rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 2: Use part f) in the definition of $\langle\!\langle\cdot\rangle\!\rangle_{c,-}$ and equation (3).

$$\begin{aligned}
&A^{-1}\langle\!\langle w_1 x_m x_c x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&\overset{f}{=} -\langle\!\langle w_1 x_m\lambda P_0(x_c)^{k'}\rangle\!\rangle_{c,-} + 2A^{-1}\langle\!\langle w_1 x_m P_1(x_c)^{k'}\rangle\!\rangle_{c,-} + A\langle\!\langle w_1 x_m x_{c-1}(x_c)^{k'+1}\rangle\!\rangle_{c,-}\\
&\overset{f}{=} -\langle\!\langle w_1 x_m\lambda P_0(x_c)^{k'}\rangle\!\rangle_{c,-} + 2A^{-1}\langle\!\langle w_1 x_m P_1(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&- A^2\langle\!\langle w_1\lambda P_{c-m}(x_c)^{k'+1}\rangle\!\rangle_{c,-} + 2A\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'+1}\rangle\!\rangle_{c,-} + A^3\langle\!\langle w_1 x_{m-1}(x_c)^{k'+2}\rangle\!\rangle_{c,-}\\
&\overset{(3)}{=} -\langle\!\langle w_1 x_m\lambda P_0(x_c)^{k'}\rangle\!\rangle_{c,-} + 2A^{-1}\langle\!\langle w_1 x_m P_1(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&- A^3\langle\!\langle w_1\lambda P_{c-m-1}(x_c)^{k'+1}\rangle\!\rangle_{c,-} + A^3\langle\!\langle w_1 x_{m-1}(x_c)^{k'+2}\rangle\!\rangle_{c,-} + A\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'+1}\rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 3: Use part f) in the definition of $\langle\!\langle\cdot\rangle\!\rangle_{c,-}$.

$$\begin{aligned}
&A\langle\!\langle w_1 x_{m-1} x_{c+1} x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&\overset{f}{=} -A^2\langle\!\langle w_1 x_{m-1}\lambda P_{-1}(x_c)^{k'}\rangle\!\rangle_{c,-} + 2A\langle\!\langle w_1 x_{m-1} P_0(x_c)^{k'}\rangle\!\rangle_{c,-} + A^3\langle\!\langle w_1 x_{m-1}(x_c)^{k'+2}\rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 4: Use equation (20).

$$\begin{aligned}
&A^{-1}\langle\!\langle w_1 P_{c-m+2} x_{c-1}(x_c)^{k'}\rangle\!\rangle_{c,-}\\
&\overset{(20)}{=} A\langle\!\langle w_1 P_{c-m+1}(x_c)^{k'+1}\rangle\!\rangle_{c,-} - A\langle\!\langle w_1 x_{m-3}(x_c)^{k'}\rangle\!\rangle_{c,-} + A^{-1}\langle\!\langle w_1 x_{m-1}(x_c)^{k'}\rangle\!\rangle_{c,-}.
\end{aligned}$$

Cancel terms, and then expand P polynomials.
Term 1: After canceling, use equation (13).

$$\begin{aligned}
&- A^3 \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-} + A \langle\!\langle w_1 x_{m+1} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{(13)}{=} -A^3 \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-} + A^2 \langle\!\langle w_1 x_m \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 2: After canceling, use definition 3.2.

$$\begin{aligned}
&- \langle\!\langle w_1 x_m \lambda P_0 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{3.2}{=} A^2 \langle\!\langle w_1 x_m \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-2} \langle\!\langle w_1 x_m \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} - 2A^2 \langle\!\langle w_1 x_m \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&= -A^2 \langle\!\langle w_1 x_m \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-2} \langle\!\langle w_1 x_m \lambda (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 3: After canceling, use definition 3.2.

$$\begin{aligned}
&- A^2 \langle\!\langle w_1 x_{m-1} \lambda P_{-1} (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A \langle\!\langle w_1 x_{m-1} P_0 (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{3.2}{=} A^{-1} \langle\!\langle w_1 x_{m-1} \lambda^2 (x_c)^{k'} \rangle\!\rangle_{c,-} - 2A^3 \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-} - 2A^{-1} \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 4: After canceling, use equation (13).

$$\begin{aligned}
&- A \langle\!\langle w_1 x_{m-3} (x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1} \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{(13)}{=} -\langle\!\langle w_1 x_{m-2} \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{(13)}{=} -A^{-1} \langle\!\langle w_1 x_{m-1} \lambda^2 (x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-2} \langle\!\langle w_1 x_m \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_{m-1} (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Now in the equation for Case iii) (Term 1 + Term 2 = Term 3 + Term 4), all the terms cancel.

For Case iv) To prove, break down each term one at a time, and then combine.
Term 1: Use equation (20) and equation (3).

$$\begin{aligned}
&A \langle\!\langle w_1 P_{c-1-m} x_{c-1} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{(20)}{=} A^3 \langle\!\langle w_1 P_{c-m-2} (x_c)^{k'+1} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} + A \langle\!\langle w_1 x_{m+2} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{(3)}{=} A^2 \langle\!\langle w_1 \lambda P_{c-m-1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} - A \langle\!\langle w_1 P_{c-m} (x_c)^{k'+1} \rangle\!\rangle_{c,-} \\
&- A^3 \langle\!\langle w_1 x_m (x_c)^{k'1} \rangle\!\rangle_{c,-} + A \langle\!\langle w_1 x_{m+2} (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 2: Use part f) in the definition of $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$, equation (12), and equation (4).

$$\begin{aligned}
&A^{-1} \langle\!\langle w_1 x_m x_{c-1} x_{c-1} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\overset{f}{=} -\langle\!\langle w_1 x_m \lambda P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m P_2 (x_c)^{k'} \rangle\!\rangle_{c,-} + A \langle\!\langle w_1 x_m x_{c-2} (x_c)^{k'+1} \rangle\!\rangle_{c,-} \\
&\overset{(12)}{=} -\langle\!\langle w_1 x_m \lambda P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m P_2 (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&+ A^2 \langle\!\langle w_1 x_m \lambda x_{c-1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
&\overset{f}{=} -\langle\!\langle w_1 x_m \lambda P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m P_2 (x_c)^{k'} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
&- A^3 \langle\!\langle w_1 \lambda P_{c-m,1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} + 2A^2 \langle\!\langle w_1 P_{c-m+1,1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} + A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
&\overset{(4)}{=} -\langle\!\langle w_1 x_m \lambda P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m P_2 (x_c)^{k'} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-} + A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
&- A^4 \langle\!\langle w_1 \lambda P_{c-m+1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} - A^2 \langle\!\langle w_1 \lambda P_{c-m-1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} \\
&+ 2A^3 \langle\!\langle w_1 P_{c-m+2} (x_c)^{k'+1} \rangle\!\rangle_{c,-} + 2A \langle\!\langle w_1 P_{c-m} (x_c)^{k'+1} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 3: Use part f) in the definition of $\langle\!\langle \cdot \rangle\!\rangle_{c,-}$.

$$\begin{aligned}
&A\langle\!\langle w_1 x_{m-1} x_c x_{c-1} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\stackrel{f}{=} -A^2 \langle\!\langle w_1 x_{m-1} \lambda P_0 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A \langle\!\langle w_1 x_{m-1} P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + A^3 \langle\!\langle w_1 x_{m-1} x_{c-1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} \\
&\stackrel{f}{=} -A^2 \langle\!\langle w_1 x_{m-1} \lambda P_0 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A \langle\!\langle w_1 x_{m-1} P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&- A^4 \langle\!\langle w_1 \lambda P_{c-m+1} (x_c)^{k'+1} \rangle\!\rangle_{c,-} + 2A^3 \langle\!\langle w_1 P_{c-m+2} (x_c)^{k'+1} \rangle\!\rangle_{c,-} + A^5 \langle\!\langle w_1 x_{m-2} (x_c)^{k'+2} \rangle\!\rangle_{c,-}
\end{aligned}$$

Term 4: Use equation (20).

$$\begin{aligned}
&A^{-1} \langle\!\langle w_1 P_{c-m+1} x_{c-1} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\stackrel{(20)}{=} A \langle\!\langle w_1 P_{c-m} (x_c)^{k'+1} \rangle\!\rangle_{c,-} - A \langle\!\langle w_1 x_{m-2} (x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1} \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Cancel terms, and then expand P polynomials and change x_m curves.
Term 1: After canceling, use equation (13).

$$\begin{aligned}
&- A^3 \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} + A \langle\!\langle w_1 x_{m+2} (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\stackrel{(13)}{=} -A^3 \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} + A^2 \langle\!\langle w_1 x_{m+1} \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\stackrel{(13)}{=} -2A^3 \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} + A^3 \langle\!\langle w_1 x_m \lambda^2 (x_c)^{k'} \rangle\!\rangle_{c,-} - A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 2: After canceling, use (3) and definition 3.2.

$$\begin{aligned}
&- \langle\!\langle w_1 x_m \lambda P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m P_2 (x_c)^{k'} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-} + A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
&\stackrel{(3)}{=} \langle\!\langle w_1 x_m \lambda P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} - 2A \langle\!\langle w_1 x_m P_0 (x_c)^{k'} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-} + A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
&\stackrel{3.2}{=} -A^3 \langle\!\langle w_1 x_m \lambda^2 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^3 \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&- A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-} + A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'+2} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 3: After canceling, use equation (13) and 3.2.

$$\begin{aligned}
& -A^2 \langle\!\langle w_1 x_{m-1} \lambda P_0 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A \langle\!\langle w_1 x_{m-1} P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + A^5 \langle\!\langle w_1 x_{m-2} (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
\stackrel{(13)}{=} \; & -A^2 \langle\!\langle w_1 x_{m-1} \lambda P_0 (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A \langle\!\langle w_1 x_{m-1} P_1 (x_c)^{k'} \rangle\!\rangle_{c,-} + A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'+2} \rangle\!\rangle_{c,-} \\
- \; & A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-} \stackrel{3.2}{=} -A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} + \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} \\
+ \; & A^4 \langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'+2} \rangle\!\rangle_{c,-} - A^3 \langle\!\langle w_1 x_m (x_c)^{k'+2} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Term 4: After canceling, use equation (13).

$$\begin{aligned}
&- A \langle\!\langle w_1 x_{m-2} (x_c)^{k'} \rangle\!\rangle_{c,-} + A^{-1} \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-} \\
&\stackrel{(13)}{=} -\langle\!\langle w_1 x_{m-1} \lambda (x_c)^{k'} \rangle\!\rangle_{c,-} + 2A^{-1} \langle\!\langle w_1 x_m (x_c)^{k'} \rangle\!\rangle_{c,-}.
\end{aligned}$$

Now in the equation for Case iv) (Term 1 + Term 2 = Term 3 + Term 4), all the terms cancel. □

Lemma 3.18. *For all $D, D' \in \mathcal{D}(\mathbf{A}^2)$ related by $\Omega_1 - \Omega_5$-moves, $h(D - D') = 0$.*

This proof mirrors proof of Lemma 3.13 from [1].

Proof. If D and D' are related by Ω_i-move with $i = 1, 2, 3, 4$, then we already know $h_1(D - D') = 0$. Thus,

$$h(D - D') = (h_3 \circ h_2)(h_1(D - D')) = (h_3 \circ h_2)(0) = 0.$$

Now we must check diagrams D and D' related by an Ω_5-move in a 2-disk $\mathbf{D}^2 \subset \mathbf{A}^2$. Move Ω_5 preserves the number of crossings, so we may label them $c_1, c_2, \ldots, c_{k-1}, c_k$ and $c_1, c_2, \ldots, c_{k-1}, c'_k$ with c_k and c'_k being the crossing on which the Ω_5-move is performed. It suffices to show h is invariant under a single Ω_5-move on one crossing as we can apply this argument inductively on all combinations of Ω-moves.

Let $\mathcal{K}(D)$ be all possible Kauffman states of D by marking each crossing as positive or negative. There are 2^k Kauffman states in $\mathcal{K}(D)$. Let D_+ and D_- be the diagram where the c_k crossing is marked positive and negative respectively. Then

$$\mathcal{K}(D) = \mathcal{K}(D_+) \cup \mathcal{K}(D_-) \quad \text{and} \quad \mathcal{K}(D') = \mathcal{K}(D'_+) \cup \mathcal{K}(D'_-).$$

Let $s \in S_{k-1} = \{-1, 1\}^{k-1}$ be a Kauffman state for all crossings except c_k and c'_k. Let s_+ (or s_-) be the same Kauffman state with a positive (or negative) marking assigned for c_k or c'_k. $\mathcal{K}(D)$ and $\mathcal{K}(D')$ contain identical states; the crossings themselves might be slightly different, but the states are the same. In particular, only the crossing c_k has been changed by the Ω_5-move to become c'_k. Smooth all crossings according to s which leaves out the c_k and c'_k crossings. As observed in [5], this results in five different cases of pairs of arrow diagrams D and D' related by a single Ω_5-move. Figures of these five cases are shown below.

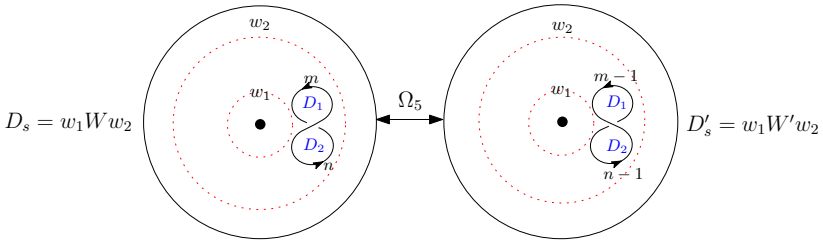


FIGURE 3.14. Case 1: Diagrams D_s and D'_s related by a Ω_5-move

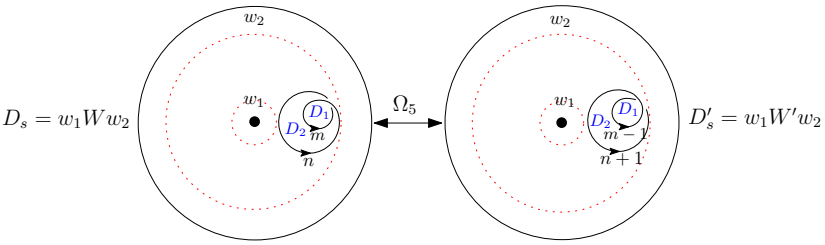


FIGURE 3.15. Case 2: Diagrams D_s and D'_s related by a Ω_5-move

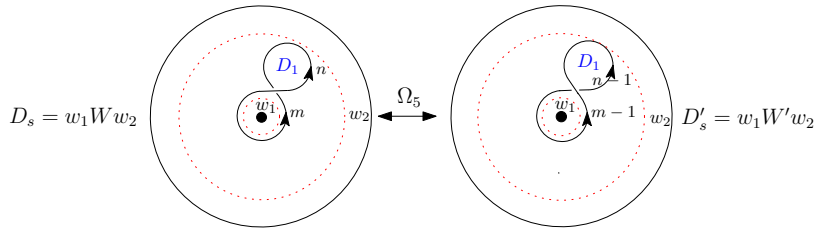


FIGURE 3.16. Case 3: Diagrams D_s and D'_s related by a Ω_5-move

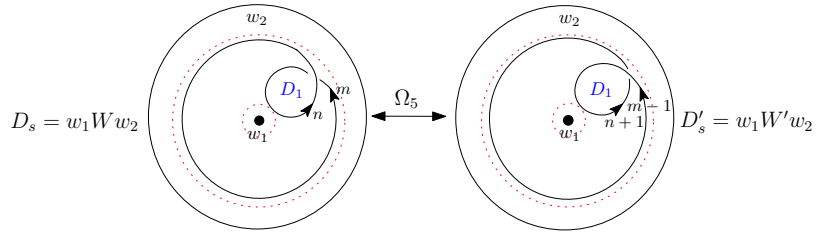


FIGURE 3.17. Case 4: Diagrams D_s and D'_s related by a Ω_5-move

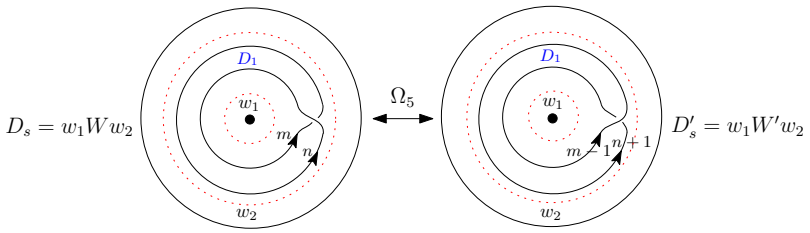


FIGURE 3.18. Case 5: Diagrams D_s and D'_s related by a Ω_5-move

Now remove all trivial circles from D_1, D_2, w_1, w_2, and let $\alpha(s)$ be the total number of trivial circles removed. Relabel D_1, D_2, w_1, w_2 as $D_1(s), D_2(s)$, $w_1(s)$, $w_2(s)$ with crossings smoothed according to s and trivial circles removed. Now smooth the final c_k or c'_k crossing in each D and D' diagram positively or negatively resulting in D_{s_+}, D'_{s_+}, D_{s_-} and D'_{s_-} for each of the five cases. The full smoothing of the diagrams according to s_+ and s_- is shown below for each of the five cases.

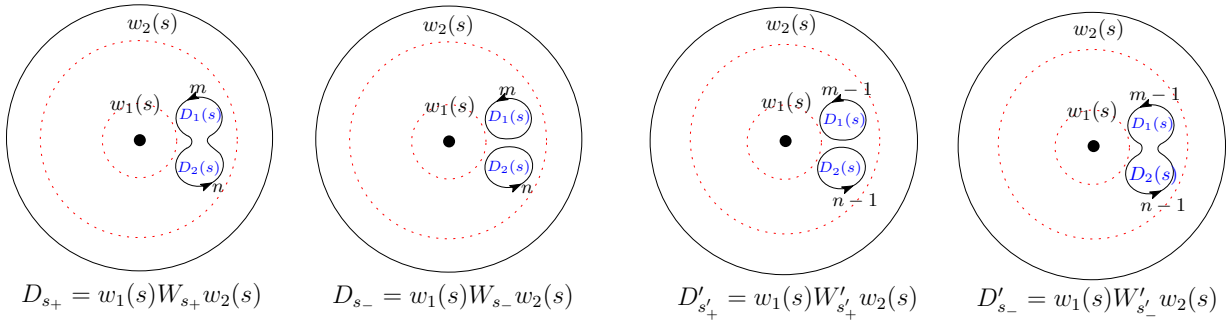


FIGURE 3.19. D_{s_+}, D'_{s_+}, D_{s_-} and D'_{s_-} from Case 1

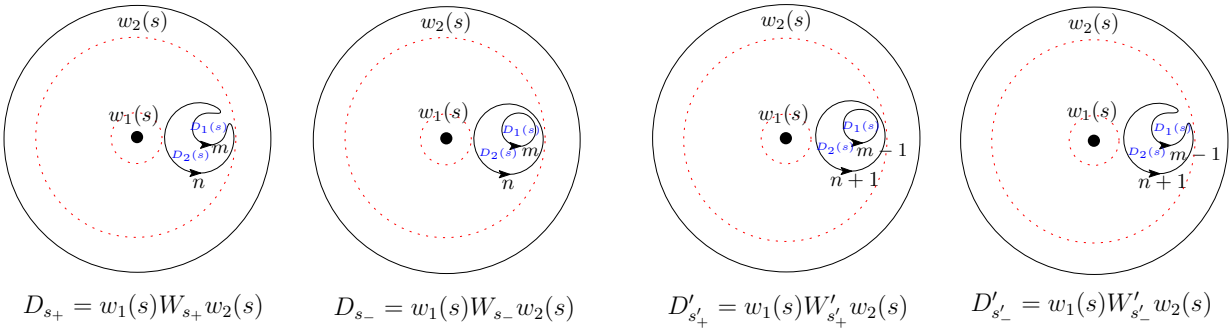


FIGURE 3.20. D_{s_+}, D'_{s_+}, D_{s_-} and D'_{s_-} from Case 2

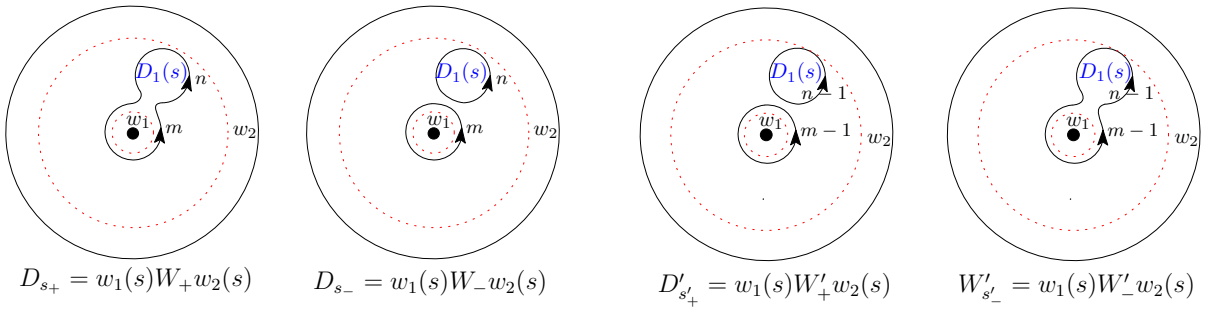


FIGURE 3.21. D_{s_+}, D'_{s_+}, D_{s_-} and D'_{s_-} from Case 3

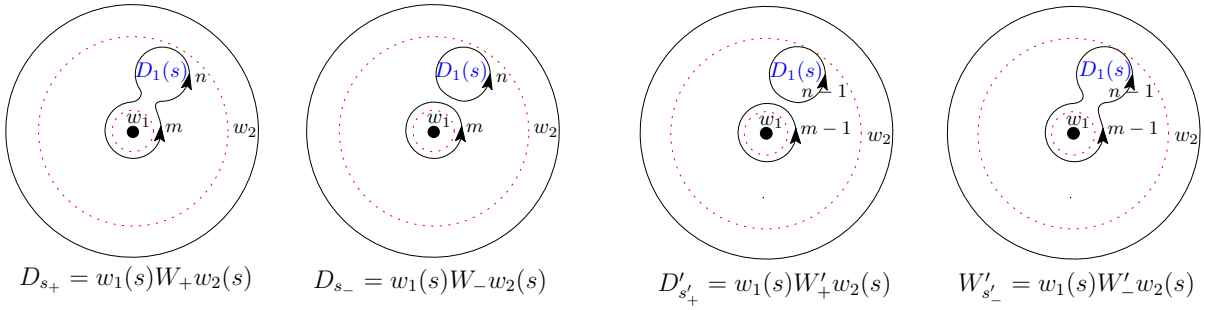


FIGURE 3.22. D_{s_+}, D'_{s_+}, D_{s_-} and D'_{s_-} from Case 4

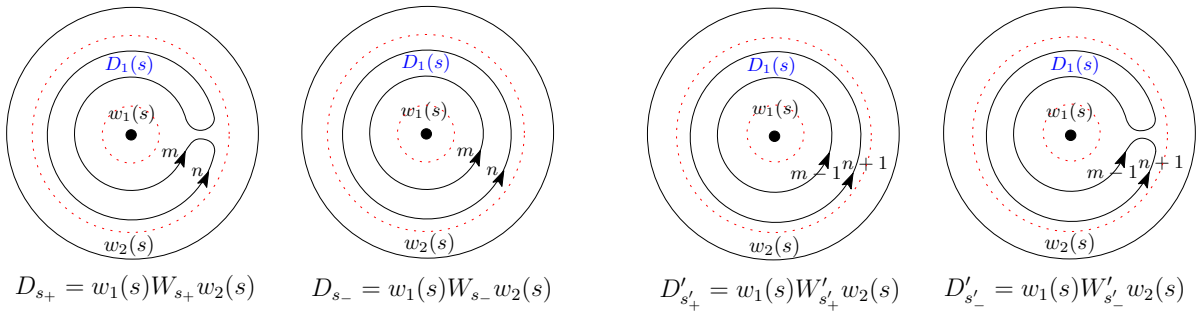


FIGURE 3.23. D_{s_+}, D'_{s_+}, D_{s_-} and D'_{s_-} from Case 5

Recall from the definition of h_1 that $p(s)$ and $n(s)$ is the number of positive and negative markers of s. Let

$$C_s(A) = A^{p(s)-n(s)}(-A^{-2}-A^2)^{\alpha(s)}.$$

Using the definition of h_1 (8) we see that,

$$h_1(D-D') \quad = \quad \sum_{s\in S_{k-1}} C_s(A)w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W'_{s'_+}\rangle - A^{-1}\langle W'_{s'_-}\rangle)w_2(s).$$

We want to show $h(D-D')=0$, and, using above, it suffices to show that

$$h_3(h_2(w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W'_{s'_+}\rangle - A^{-1}\langle W'_{s'_-}\rangle)w_2(s))) = 0.$$

In Case 1 and Case 2, the curves on which the the crossing c_k and c'_k crosses are completely contained in a 2-disk $\mathbf{D}^2 \subset \mathbf{A}^2$. In these cases, we can use the invariance of $D \in \mathcal{D}(\mathbf{D}^2)$ under $\Omega_1 - \Omega_5$-moves on $\mathbf{D}^2$.

As in Figure 3.14 and 3.15, let $D = w_1 W w_2$ and $D' = w_1 W' w_2$. Let W_{s_+}, W_{s_-}, $W'_{s'_+}$, and $W'_{s'_-}$ be the diagrams with the crossing c_k or c'_k smoothed positively or negatively.

$$\begin{aligned} h_2(h_1(D - D')) &= h_2(w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W'_{s'_+}\rangle - A^{-1}\langle W'_{s'_-}\rangle)w_2(s)) \\ &= \langle\!\langle w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W'_{s'_+}\rangle - A^{-1}\langle W'_{s'_-}\rangle)w_2(s)\rangle\!\rangle_\Gamma \\ &= \langle\!\langle w_1(s)\rangle\!\rangle_\Gamma(\langle AW_{s_+} + A^{-1}W_{s_-} - AW'_{s'_+} - A^{-1}W'_{s'_-}\rangle_r)\langle\!\langle w_2(s)\rangle\!\rangle_\Gamma \\ &= \langle\!\langle w_1(s)\rangle\!\rangle_\Gamma(A\langle W_{s_+}\rangle_r + A^{-1}\langle W_{s_-}\rangle_r - A\langle W'_{s'_+}\rangle_r - A^{-1}\langle W'_{s'_-}\rangle_r)\langle\!\langle w_2(s)\rangle\!\rangle_\Gamma \\ &= \langle\!\langle w_1(s)\rangle\!\rangle_\Gamma(\langle W\rangle_r - \langle W'\rangle_r)\langle\!\langle w_2(s)\rangle\!\rangle_\Gamma \end{aligned}$$

We know $\langle\cdot\rangle_r$ is invariant under $\Omega_1 - \Omega_5$-moves on $\mathbf{D}^2$, so $\langle W\rangle_r = \langle W'\rangle_r$. Therefore,

$$h_2(w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W_{s'_+}\rangle - A\langle W_{s'_-}\rangle)w_2(s)) = 0. \tag{30}$$

Use equation (30) for Figure 3.14 and 3.15.

$$\begin{aligned} & h(D - D') \\ = \; & (h_3 \circ h_2)(w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W_{s'_+}\rangle - A\langle W_{s'_-}\rangle)w_2(s)) = 0. \end{aligned}$$

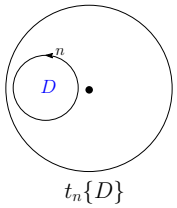


FIGURE 3.24. Diagram $t_n\{D\}$

Let $t_n\{D\}$ be the arrow diagram consisting of an arrow diagram D inside t_n (see Figure 3.24). Let

$$\langle D_1(s)\rangle_r = \sum_{i_1=0}^{k_1} r_{1,i_1}\lambda^{i_1}, \quad \langle\!\langle w_1(s)\rangle\!\rangle_\Gamma = \sum_{i_2=0}^{k_2} r_{2,i_2}u_{2,i_2}, \quad \text{and} \quad \langle\!\langle w_2(s)\rangle\!\rangle_\Gamma = \sum_{i_3=0}^{k_3} r_{3,i_3}u_{3,i_3}.$$

For Case 3: In Figure 3.21

$$W_{s_+} = D_1(s)x_{n+m} \qquad W_{s_-} = x_m t_n\{D_1(s)\}$$
$$W'_{s'_+} = x_{m-1}t_{n-1}\{D_1(s)\} \quad W'_{s'_-} = D_1(s)x_{n+m-2}$$

For Case 4: In Figure 3.22

$$W_{s_+} = t_n\{D_1(s)\}x_m \qquad W_{s_-} = x_{m-n}D_1(s)$$
$$W'_{s'_+} = x_{m-n-2}D_1(s) \quad W'_{s'_-} = t_{n+1}\{D_1(s)\}x_{m-1}$$

For Case 5: In Figure 3.23

$$W_{s_+} = t_{n-m}\{D_1(s)\} \qquad W_{s_-} = x_m D_1(s)x_n$$
$$W'_{s'_+} = x_{m-1}D_1(s)x_{n+1} \quad W'_{s'_-} = t_{n-m+2}\{D_1(s)\}$$

Evaluate $h(D - D') = 0$ for D and D' in Figure 3.21, Figure 3.22, and Figure 3.23.

For Figure 3.21

$$\begin{aligned}
& h(D-D') \\
= \;& (h_3 \circ h_2)(w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W_{s'_+}\rangle - A\langle W_{s'_-}\rangle)w_2(s)) \\
= \;& h_3(\langle\!\langle (w_1(s)(A\langle W_{s_+}\rangle + A^{-1}\langle W_{s_-}\rangle - A\langle W_{s'_+}\rangle - A\langle W_{s'_-}\rangle)w_2(s)\rangle\!\rangle_\Gamma) \\
= \;& \sum_{i_1=0}^{k_1} r_{1,i_1} h_3(\langle\!\langle w_1(s)\rangle\!\rangle_\Gamma (A\lambda^{i_1} x_{n+m} + A^{-1} x_m P_{n,i_1} - A x_{m-1} P_{n-1,i_1} - A\lambda^{i_1} x_{n+m-2})\langle\!\langle w_2(s)\rangle\!\rangle_\Gamma) \\
= \;& \sum_{i_1=0}^{k_1}\sum_{i_2=0}^{k_2}\sum_{i_3=0}^{k_3} r_{1,i_1} r_{2,i_2} r_{3,i_3} h_3(u_{2,i_2}(A\lambda^{i_1} x_{n+m} + A^{-1} x_m P_{n,i_1} - A x_{m-1} P_{n-1,i_1} - A\lambda^{i_1} x_{n+m-2})u_{3,i_3}).
\end{aligned}$$

This is equal to zero by Lemma 3.15, that is, equation (21).

For Figure 3.22

$$\begin{aligned}
& h(D-D') \\
= \;& \sum_{i_1=0}^{k_1}\sum_{i_2=0}^{k_2}\sum_{i_3=0}^{k_3} r_{1,i_1} r_{2,i_2} r_{3,i_3} h_3(u_{2,i_2}(A P_{n,i_1} x_m + A^{-1} x_{m-n}\lambda^{i_1} - A x_{m-n-2}\lambda^{i_1} - A P_{n+1,i_1} x_{m-1})u_{3,i_3}).
\end{aligned}$$

This is equal to zero by Lemma 3.16, that is, equation (24).

For Figure 3.23

$$\begin{aligned}
& h(D-D') \\
= \;& \sum_{i_1=0}^{k_1}\sum_{i_2=0}^{k_2}\sum_{i_3=0}^{k_3} r_{1,i_1} r_{2,i_2} r_{3,i_3} h_3(u_{2,i_2}(A P_{n-m,i_1} + A^{-1} x_m \lambda^{i_1} x_n - A x_{m-1}\lambda^{i_1} x_{n+1} - A P_{n-m+2,i_1})u_{3,i_3}).
\end{aligned}$$

This is equal to zero by Lemma 3.17, that is, equation (27). This exhausts all cases for pairs of arrow diagrams D and D' that we need to check. Consequently, $h(D-D')=0$ for D and D' related by $\Omega_1-\Omega_5$-moves. □

Let $\overline{\Sigma}_{c,-}$ be the set of equivalence classes of arrow diagrams in $\Sigma_{c,-}$ modulo $\Omega_1-\Omega_5$-moves. To end this section, following [1], we state analogous result for basis $\Sigma_{c,-}$.

Theorem 3.19. *KBSM of $\mathbf{A}^2 \times S^1$ is a free R-module with basis that consists of ambient isotopy classes of generic framed links whose arrow diagrams are in $\overline{\Sigma}_{c,-}$, i.e.,*

$$\mathcal{S}_{2,\infty}(\mathbf{A}^2 \times S^1; R, A) \cong R\overline{\Sigma}_{c,-}.$$

Proof. Lemma 3.18 shows our h is invariant under $\Omega_1-\Omega_5$ moves. All other definitions and conclusions are identical to [1]. Therefore the proof of Theorem 3.19 follows the proof of Theorem 3.14 from [1] exactly. □

Since $c \in \mathbb{Z}$ is arbitrary, Theorem 3.19 gives an infinite family of bases which are different than those found in [7], [5], and [1].

4. Computational Basis Transfer Algorithm

Computing h to express an arrow diagram D on $\mathbf{A}^2$ as an R-linear combination of $\Sigma_{c,-}$ can become computationally difficult even for small link diagrams. To aid in this, a computer algorithm implemented in Python can be used to compute h_3 for several different values of c. Recall, h_3 takes in elements from $R\Gamma$, and it returns R-linear combinations in basis $\Sigma_{c,-}$. The script can replicate h_3 for basis $\Sigma_{c,-}$ as well as h_3 for basis Σ_c from [1].

$$\begin{aligned}
\Sigma_c &= \{\lambda^n, x_c\lambda^n(x_c)^k, x_{c+1}\lambda^n(x_c)^k \mid \; n,k \geq 0\}, \\
\Sigma_{c,-} &= \{\lambda^n, x_c\lambda^n(x_c)^k, x_{c-1}\lambda^n(x_c)^k \mid \; n,k \geq 0\}.
\end{aligned}$$

It can also transfer elements to conjectured bases $\Sigma_{c,+,b}$ and $\Sigma_{c,-,b}$ defined as follows: For $c, b \in \mathbb{Z}$ and $b \geq 0$,

$$\begin{aligned}
\Sigma_{c,+,b} = \{&\lambda^n, x_c\lambda^n, x_{c+1}x_c\lambda^n, ..., (x_{c+1})^{b-1}x_c\lambda^n, \\
&x_{c+1}\lambda^n, x_{c+1}x_{c+1}\lambda^n, ..., (x_{c+1})^{b-1}x_{c+1}\lambda^n, \\
&(x_{c+1})^b x_c\lambda^n(x_c)^k, (x_{c+1})^b x_{c+1}\lambda^n(x_c)^k \mid \; n,k \geq 0\}.
\end{aligned}$$

$$\begin{aligned}\Sigma_{c,-,b} = \{&\lambda^n, x_c\lambda^n, x_{c-1}x_c\lambda^n, ..., (x_{c-1})^{b-1}x_c\lambda^n,\\ &x_{c-1}\lambda^n, x_{c-1}x_{c-1}\lambda^n, ..., (x_{c-1})^{b-1}x_{c-1}\lambda^n,\\ &(x_{c-1})^b x_c\lambda^n(x_c)^k, (x_{c-1})^b x_{c-1}\lambda^n(x_c)^k \mid \ n,k\geq 0\}.\end{aligned}$$

Notice with $b=0$, the basis $\Sigma_{c,+,0}$ is Σ_c and $\Sigma_{c,-,0}$ is $\Sigma_{c,-}$. The algorithm for h_3 becomes longer for larger b, but an algorithm that transfers from $\Sigma_{c,+,b}$ to $\Sigma_{c,+,b+1}$ is fixed in length. To write any element w of $R\Gamma$ as a linear combination of basis elements from $\Sigma_{c,+,b}$, first the algorithm writes w in $\Sigma_{c,+,0}$. Then it transfers up to $\Sigma_{c,+,b}$ by adding one to b on each step.

For $b\geq 0$, the basis transfer algorithm from $R\Sigma_{c,+,b}$ to $R\Sigma_{c,+,b+1}$ is given by the following algorithm.

Definition 4.1. Let $u\in R\Sigma_{c,+,b}$ with $b\geq 0$. We define $\langle\!\langle u\rangle\!\rangle_{c,+,b+1}\in R\Sigma_{c,+,b+1}$ as follows:

(a) if $u=\sum_{w\in S} r_w w$, S is a finite subset of $R\Sigma_{c,+,b}$ with at least two elements, let
$$\langle\!\langle u\rangle\!\rangle_{c,+,b+1} = \sum_{w\in S} r_w \langle\!\langle w\rangle\!\rangle_{c,+,b+1};$$

(b) for $w\in\Sigma_{c,+,b+1}$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = w;$$

(c) for $w=w'\lambda^0 x_{c+1}\lambda^n x_m(x_c)^k\in\Gamma$ with w' containing at least b amount of x curves and $n\geq 1$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A\langle\!\langle w'\lambda^0 x_{c+1}\lambda^{n-1}x_{m+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} + A^{-1}\langle\!\langle w'\lambda^0 x_{c+1}\lambda^{n-1}x_{m-1}(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(d) for $w=w'\lambda^0 x_{c+1}x_m\lambda^n(x_c)^k\in\Gamma$ with w' containing at least b amount of x curves and $m>c+1$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A\langle\!\langle w'\lambda^0 x_{c+1}x_{m-1}\lambda^{n+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} - A^2\langle\!\langle w'\lambda^0 x_{c+1}x_{m-2}\lambda^n(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(e) for $w=w'\lambda^0 x_{c+1}x_m\lambda^n(x_c)^k\in\Gamma$ with w' containing at least b amount of x curves and $m<c$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A^{-1}\langle\!\langle w'\lambda^0 x_{c+1}x_{m+1}\lambda^{n+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} - A^{-2}\langle\!\langle w'\lambda^0 x_{c+1}x_{m+2}\lambda^n(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(f) for $w=w'\lambda^0 x_c\lambda^n x_c(x_c)^k\in\Gamma$ with w' containing at least b amount of x curves,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = -A^{-1}\langle\!\langle w'\lambda P_{-1,n}(x_c)^k\rangle\!\rangle_{c,+,b+1}+2\langle\!\langle w'\lambda^0 P_{-2,n}(x_c)^k\rangle\!\rangle_{c,+,b+1}+A^{-2}\langle\!\langle w'\lambda^0 x_{c+1}\lambda^n x_{c-1}(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(g) for $w=w'x_{c+1}\lambda^n x_m(x_c)^k\in\Gamma$ with $n\geq 1$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A\langle\!\langle w'x_{c+1}\lambda^{n-1}x_{m+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} + A^{-1}\langle\!\langle w'x_{c+1}\lambda^{n-1}x_{m-1}(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(h) for $w=w'x_{c+1}x_m\lambda^n(x_c)^k\in\Gamma$ with $m>c+1$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A\langle\!\langle w'x_{c+1}x_{m-1}\lambda^{n+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} - A^2\langle\!\langle w'x_{c+1}x_{m-2}\lambda^n(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(i) for $w=w'x_{c+1}x_m\lambda^n(x_c)^k\in\Gamma$ with $m<c$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A^{-1}\langle\!\langle w'x_{c+1}x_{m+1}\lambda^{n+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} - A^{-2}\langle\!\langle w'x_{c+1}x_{m+2}\lambda^n(x_c)^k\rangle\!\rangle_{c,+,b+1}.$$

(j) for $w=\lambda^n x_m(x_c)^k\in\Gamma$ with $n\geq 1$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A\langle\!\langle \lambda^{n-1}x_{m+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} + A^{-1}\langle\!\langle \lambda^{n-1}x_{m-1}(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(k) for $w=x_m\lambda^n(x_c)^k\in\Gamma$ with $m>c+1$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A\langle\!\langle x_{m-1}\lambda^{n+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} - A^2\langle\!\langle x_{m-2}\lambda^n(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(l) for $w=x_m\lambda^n(x_c)^k\in\Gamma$ with $m<c$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A^{-1}\langle\!\langle x_{m+1}\lambda^{n+1}(x_c)^k\rangle\!\rangle_{c,+,b+1} + A^{-2}\langle\!\langle x_{m+2}\lambda^n(x_c)^k\rangle\!\rangle_{c,+,b+1};$$

(m) for $w=x_m\lambda^n\in\Gamma$ with $m>c+1$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A\langle\!\langle x_{m-1}\lambda^{n+1}\rangle\!\rangle_{c,+,b+1} + A^2\langle\!\langle x_{m-2}\lambda^n\rangle\!\rangle_{c,+,b+1};$$

(n) for $w=x_m\lambda^n\in\Gamma$ with $m<c$,
$$\langle\!\langle w\rangle\!\rangle_{c,+,b+1} = A^{-1}\langle\!\langle x_{m+1}\lambda^{n+1}\rangle\!\rangle_{c,+,b+1} + A^{-2}\langle\!\langle x_{m+2}\lambda^n\rangle\!\rangle_{c,+,b+1}.$$

All parts, except (f), follow from equation (10). Part (f) follows from equation (8) in [1]. To write any element w of $R\Gamma$ as an R-linear combination of $\Sigma_{c,-,b}$, follow the same process of writing w in $\Sigma_{c,-,0}$ and then transferring up to $\Sigma_{c,-,b}$ by adding one to b on each step. The basis transfer algorithm from $R\Sigma_{c,-,b}$ to $R\Sigma_{c,-,b+1}$ is similar. The main difference is part (f) which uses equation (11) instead of equation (8) in [1].

Example 4.2. We will show a sample calculation for $rw = Ax_5x_5x_5x_4\lambda x_4 \in R\Gamma$. rw is an element of $R\Sigma_{4,+,3}$. We will use the algorithm to write rw in $R\Sigma_{4,+,0}$. Then use the transfer algorithm to write rw in $R\Sigma_{4,+,1}$, $R\Sigma_{4,+,2}$, and $R\Sigma_{4,+,3}$ where rw should return to $Ax_5x_5x_5x_4\lambda x_4$.

For $rw \in R\Sigma_{4,+,0}$:

$$\begin{aligned}
&\left(A^8 - 12A^4 + 11 - 4A^{-4} + 2A^{-8} + 3A^{-12} - A^{-16} - 2A^{-20} + 3A^{-24} - A^{-28}\right) x_4 \\
&+ \left(7A^4 - 5 + 2A^{-4} - 2A^{-8} - 2A^{-12} + A^{-16} - 2A^{-24} + A^{-28}\right) (x_4)^3 \\
&+ \left(-5A^3 + 4A^{-1} + 3A^{-9} - A^{-13} + A^{-17} - A^{-21} - 4A^{-25} + 3A^{-29}\right) x_5\lambda(x_4)^2 \\
&+ \left(A^{-1} + 3A^{-9}\right) x_5\lambda^2(x_4)^2 + \left(-A^{-9}\right) x_5\lambda^4(x_4)^2 \\
&+ \left(A^3 - 2A^{-9}\right) x_5\lambda(x_4)^4 + \left(-A^4 + A^{-8}\right) (x_4)^5 \\
&+ \left(-2A^7 + 7A^3 - 4A^{-1} - 3A^{-5} + A^{-9} + 2A^{-13} - 2A^{-17} - 3A^{-21} + 7A^{-25} - 3A^{-29}\right) x_5\lambda \\
&+ \left(-A^8 + A^4 - 2 + 9A^{-4} - 8A^{-8} + 3A^{-16} + 4A^{-20} - 12A^{-24} + 6A^{-28}\right) x_4\lambda^2 \\
&+ \left(-A^4 - 4A^{-4} + 5A^{-8} - A^{-12} - A^{-16} + 2A^{-20} + 6A^{-24} - 6A^{-28}\right) x_4\lambda^2(x_4)^2 \\
&+ \left(A^3 + 4A^{-5} - 3A^{-9} - A^{-13} + A^{-21} + 2A^{-25} - 4A^{-29}\right) x_5\lambda^3(x_4)^2 \\
&+ \left(A^{-9}\right) x_5\lambda^3(x_4)^4 + \left(-A^{-8}\right) x_4\lambda^2(x_4)^4 \\
&+ \left(A^7 - A^3 + 3A^{-1} - 5A^{-5} + A^{-9} + A^{-13} + A^{-17} + A^{-21} - 6A^{-25} + 4A^{-29}\right) x_5\lambda^3 \\
&+ \left(1 - A^{-4} + A^{-8} - A^{-12} - A^{-16} - A^{-20} + 7A^{-24} - 5A^{-28}\right) x_4\lambda^4 \\
&+ \left(A^{-4} + A^{-12} - A^{-20} - 2A^{-24} + 5A^{-28}\right) x_4\lambda^4(x_4)^2 \\
&+ \left(-A^{-1} + A^{-5} + A^{-25} - A^{-29}\right) x_5\lambda^5 \\
&+ \left(-A^{-24} + A^{-28}\right) x_4\lambda^6 + \left(-A^{-5} + A^{-29}\right) x_5\lambda^5(x_4)^2 \\
&+ \left(-A^{-28}\right) x_4\lambda^6(x_4)^2.
\end{aligned}$$

For $rw \in R\Sigma_{4,+,1}$:

$$\begin{aligned}
&\left(-2A^4 + 2A^{-4}\right) x_5x_5x_4 + \left(A^4 - A^{-4}\right) x_5x_5(x_4)^3 \\
&+ \left(-A^3 + A^{-1} + A^{-13} - A^{-17}\right) x_5x_4x_4 \\
&+ \left(A^2 + 2A^{-2} - 2A^{-6} + A^{-14} - 2A^{-18}\right) x_5x_5\lambda x_4 \\
&+ \left(A^{-1} + A^{-5} - A^{-13} + 3A^{-17}\right) x_5x_4\lambda^2x_4 \\
&+ \left(1 - A^{-4}\right) x_5x_5\lambda^2x_4 + \left(A^{-4}\right) x_5x_5\lambda^2(x_4)^3 \\
&+ \left(-A^{-3}\right) x_5x_4\lambda(x_4)^3 + \left(-A^{-2} + A^{-18}\right) x_5x_5\lambda^3x_4 \\
&+ \left(-A^{-17}\right) x_5x_4\lambda^4x_4.
\end{aligned}$$

For $rw \in R\Sigma_{4,+,2}$:

$$\begin{aligned}
&\left(-3A^4 + 3 - A^{-4} + A^{-8}\right) x_5x_5x_4 \\
&+ \left(A^4 - 1\right) x_5x_5x_4(x_4)^2 \\
&+ \left(A^3 - A^{-1} - A^{-5} + A^{-9}\right) x_5x_5x_5\lambda \\
&+ \left(A^{-1}\right) x_5x_5x_5\lambda(x_4)^2 \\
&+ \left(A^{-4} - A^{-8}\right) x_5x_5x_4\lambda^2.
\end{aligned}$$

For $rw \in R\Sigma_{4,+,3}$:

$$(A)x_5x_5x_5x_4\lambda x_4.$$

In the final step, we see rw return to its form in $R\Sigma_{4,+,3}$. Even for a small word such as rw, this calculation is too tedious and prone to mistakes to perform by hand. When you factor the coefficients, some familiar terms appear.

For example $rw \in R\Sigma_{4,+,2}$ factored is

$$\begin{aligned}
&- A^{-6}(A^2 - A^{-2})(3A^8 + 1)x_5x_5x_4 \\
&+ A^2(A^2 - A^{-2})x_5x_5x_4(x_4)^2 \\
&- A^{-3}(A^2 - A^{-2})^2(-A^2 - A^{-2})x_5x_5x_5\lambda \\
&+ (A^{-1})x_5x_5x_5\lambda(x_4)^2 \\
&+ A^{-6}(A^2 - A^{-2})x_5x_5x_4\lambda^2.
\end{aligned}$$

These coefficients store information about the original link.

To check if words match the parts of the basis transfer algorithm and compute the change, we need a way to store the data of an element of $R\Gamma$. A Python dictionary is a good choice for two reasons. First, the addition in $R\Gamma$ is commutative so the set of words does not need to be ordered. Second, each word has a coefficient that we only care about accessing for that specific word. This makes a key-value pair structure good for storing word-coefficient data. The fast lookup of Python dictionaries—$O(1)$ time—is desirable because every part of the definition of $\langle\!\langle\cdot\rangle\!\rangle_{c,+,b+1}$ needs to check if the changed word is already in the linear combination.

To store the data of the coefficient, the library SymPy allows us to store elements of $\mathbb{Z}[A^{\pm 1}]$ and do algebra with them. A sorted array is needed to store the data for a word because the order matters. For the program, it made more sense to store the word in reverse order: from outside-in. For a word $w = \lambda^{n_0}x_{m_1}\lambda^{n_1}\cdots\lambda^{n_{i-1}}x_{m_i}\lambda^{n_i}(x_c)^k \in \Gamma$, store in an array as follows:

$$[c, k, n_i, m_i, n_{i-1}, ..., n_1, m_1, n_0]$$

Notice, for a fixed c, the same word can have $K+1$ different representations as an array where K is the maximum k can be.

Example 4.3. As an example, $w = x_5x_5x_5x_4\lambda x_4$ is shown as an array.

For $c = 4$ and $x_5x_5x_5x_4\lambda(x_4)^1$:

$$[4, 1, 1, 4, 0, 5, 0, 5, 0, 5, 0]$$

or $c = 4$ and $x_5x_5x_5x_4\lambda x_4(x_4)^0$:

$$[4, 0, 0, 4, 1, 4, 0, 5, 0, 5, 0, 5, 0]$$

or $c \neq 4$ and $x_5x_5x_5x_4\lambda x_4(x_c)^0$:

$$[c, 0, 0, 4, 1, 4, 0, 5, 0, 5, 0, 5, 0]$$

The canonical representation for fixed c of a word is the array with the maximum k value: K.

These words are then matched and changed according to the algorithm until every word in the linear combination is a basis element. The matching could have been done with regex on strings that look like $\lambda^{n_0}x_{m_1}\lambda^{n_1}\cdots\lambda^{n_{i-1}}x_{m_i}\lambda^{n_i}(x_c)^k$ already, but then the changes that $\langle\!\langle\cdot\rangle\!\rangle_{c,+,b+1}$ makes would have been more difficult to code. Creating a match pattern algorithm for this specific context proved more efficient.

For ease of use, a graphic user interface allows you to enter elements in LaTeX. Then the output may be copied to the clipboard in LaTeX also. This script was written as part of my participation in the 2024 Research Experience for Undergraduates at the University of Texas and Dallas. The code is available upon request or through GitHub[3].

5. Future Work

In this paper, we proved that $\Sigma_{c,-}$ is a basis of the KBSM of $\mathbf{A}^2 \times S^1$, for any $c \in \mathbb{Z}$. In the future, this result could be used to construct bases for the KBSM of fibered tori $V(\beta, 3)$ and $V(\beta, 4)$. These bases, in turn, could help compute the KBSM for families of lens spaces and small Seifert fibered 3-manifolds. For the computer program, we hope to create a web app with a better user interface design.

[3]Hayden Siebers, Basis Transfer For KBSM of $\mathbf{A}^2 \times S^1$, 2024, https://github.com/HeyHeyHayHay/KBSM-Fibered-Torus-Algorithms

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